

A CATALOGUE OF
TECHNICAL AND
SCIENTIFIC BOOKS

FORWARDED BY

CONSTABLE
& COMPANY LTD

A CATALOGUE OF
TECHNICAL AND
SCIENTIFIC BOOKS

PUBLISHED BY

CONSTABLE
& COMPANY LTD

The Project Gutenberg eBook of A catalogue of technical and scientific books published by Constable & Company Ltd

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A catalogue of technical and scientific books published by Constable & Company Ltd

Creator: Constable

Release date: March 3, 2018 [eBook #56674]

Language: English

Other information and formats: www.gutenberg.org/ebooks/56674

Credits: Produced by deaurider and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK A CATALOGUE
OF TECHNICAL AND SCIENTIFIC BOOKS PUBLISHED BY
CONSTABLE & COMPANY LTD ***

A CATALOGUE OF

TECHNICAL AND

SCIENTIFIC BOOKS

PUBLISHED BY

CONSTABLE

& COMPANY LTD

- [19](#) Aeronautics
- [58](#) Agriculture, Farming and Forestry
- [55](#) Arts and Crafts
- [55](#) Business and Management
- [27](#) Civil Engineering, Building Construction, etc.
- [33](#) Electrical Engineering
- [7](#) Electro-Chemistry
- [13](#) Fuel and its Combustion
- [21](#) Hydraulics and Turbines
- [10](#) Internal Combustion Engines
- [17](#) Iron, Steel, and other Metals
- [31](#) Irrigation and Water Supply
- [61](#) Law, Patents, etc.
- [39](#) Lighting
- [15](#) Machinery, Power Plants, etc.
- [51](#) Manufacture and Industries
- [20](#) Marine Engineering
- [50](#) Mathematics

[49](#) Metallurgy
[23](#) Mining
[61](#) Miscellaneous
[19](#) Motor-Cars and Engines
[30](#) Municipal Engineering
[57](#) Natural History, Botany, Nature Study, etc.
[8](#) Outlines of Industrial Chemistry
[40](#) Physics and Chemistry
[17](#) Pumps
[24](#) Railway Engineering
[26](#) Reinforced Concrete and Cement
[12](#) Steam Engines, Boilers, etc.
[29](#) Surveying
[32](#) Telegraphy and Telephony
[39](#) Thermodynamics
[56](#) Useful Handbooks and Tables

VARIOUS SERIES

- [7](#) A Treatise of Electro-Chemistry
- [9](#) The D.-S. Technical Dictionaries in Six Languages
- [4](#) The Glasgow Text-Books of Civil Engineering
- [65](#) Messrs. Franklin & Charles' List
- [9](#) Metallurgical Series
- [8](#) Outlines of Industrial Chemistry
- [7](#) Practical Manuals for Practical Men
- [3](#) Staple Trades and Industries
- [3](#) The Chadwick Library
- [4](#) "The Engineer" Library
- [5](#) The "Westminster" Series
- [8](#) Thresholds of Science
- [66](#) Index

LONDON
10 & 12 ORANGE STREET
LEICESTER SQUARE, W.C.2
1921

MESSRS. CONSTABLE & CO., LTD., are the European, Australian and South African Agents for publications of Messrs. D. Van Nostrand, New York, and of Messrs. Franklin & Charles of Bethlehem, Pa.

NOTE.

The cost of postage is to be added to all net prices of books in this list. Where the cost of postage is not stated, 6d. on books published at 10/- or less, and 9d. on books published at more than 10/- will generally cover the inland postal charges.

[Owing to the fluctuating costs of material and manufacture the published price of books is liable to be changed without notice.]

The Chadwick Library

Altitude and Health. **Prof. F. F. Roget.** Demy 8vo. 12/6 net.

Rainfall, Reservoirs and Water Supply. **Sir Alexander Binnie.** 9/- net. ([See p. 30.](#))

Health in Camp. **Austin T. Nankivell, M.D., Capt. R.A.M.C.,** with an Introduction by **Sir Shirley Murphy, F.R.C.S., R.A.M.C. (T.)**. Paper, 1/- net; Cloth, 1/6 net.

The Future Citizen and his Mother. **Charles Porter, M.D., B.Sc.,** with a Foreword by **Sir James Crichton Browne, M.D., D.Sc.** 3/6 net.

Forests, Woods and Trees in relation to Hygiene. **Augustine Henry, M.A., F.L.S.,** etc. Price, 18/- net. ([See p. 60.](#))

The Colloids in Health and Disease. **A. B. Searle.** Demy 8vo. 8/- net.

Domestic Fuel Consumption. **A. H. Barker, B.A., B.Sc.,** Author of "The Theory and Practice of Heating and Ventilation," etc. Demy 8vo. 180 pp. 14/- net.

Written so as to be readily comprehensible by any person without scientific training, this book will be read with interest and profit by all who, whether for private or public purposes, are desirous of checking the present lavish waste of our fuel resources.

Staple Trades and Industries

Edited by **GORDON D. KNOX**. Demy 8vo. Illustrated.

First Volumes now Ready.

***Wool.* Frank Ormerod.** 16 Illustrations. 6/6 net.

CONTENTS.—The Genesis of Wool. The Coming of the Weavers. How England Took the Lead. Sheep and Wool Fibre. The World's Wool Supply. Australia's Pastoral Romance. The Marketing of Wool. Shearing and Sorting. Preparation and Manufacture. Finishing Processes. Extremes of Woollen Production. Varied Uses. The Future. Salient Facts and Figures.

***Cotton.* George Bigwood.** 16 Illustrations. 6/6 net.

CONTENTS.—History of the Cotton Plant. The Development of Spinning. The Cotton Fields. Triumph of Mechanical Invention. Cotton Growing under the British Flag. Classification of the World's Crop. Modern Spinning and Weaving. "Where Merchants Most do Congregate." Gambling in Cotton. Cotton Fabrics; An Art Manufacture. Organisation and Strikes. A General Utility Plant. Appendices: The Cotton Trade in War Time; Spindles and Looms; "Futures."

***Linen.* Alfred S. Moore.** (In Preparation.)

***The Nitrogen Industry.* J. M. Partington, M.Sc.** (do.)

The Glasgow Text-Books of Civil Engineering

Edited by **G. MONCUR, B.Sc., M.Inst.C.E.**, Professor of Civil Engineering at the Royal Technical College, Glasgow.

Railway Signal Engineering (Mechanical). **Leonard P. Lewis**, of the Caledonian Railway; Lecturer on Railway Signalling at the Royal Technical College, Glasgow. Illustrated. Demy 8vo. 2nd Edition. 14/- net. ([See p. 25.](#))

Modern Sanitary Engineering. **Gilbert Thomson, M.A., F.R.S.E., M.Inst.C.E.** Part I. **House Drainage.** Illustrated. Demy 8vo. 2nd Edition. 16/- net. ([See p. 31.](#)) Part II. **Sewerage.** 18/- net.

Reinforced Concrete Railway Structures. **J. D. W. Ball, A.M.Inst.C.E.** Illustrated. Demy 8vo. 9/- net. (p. 25.)

Surveying and Field Work. **J. Williamson, A.M.INST.C.E.** Demy 8vo. 9/- net. ([See p. 29.](#))

Hydraulics of Pipe Lines. **W. F. Durand**, Professor of Mechanical Engineering, Leland Stanford University, California. Demy 8vo.

Earthwork in Railway Engineering. **J. W. P. Gardner, M.I.C.E.** Demy 8vo. Illustrated.

Dock and Lock Machinery. **William Hunter, M.Inst.C.E.** Demy 8vo. Illustrated.

Other Volumes in Preparation.

“The Engineer” Library

What Industry Owes to Chemical Science. **Richard B. Pilcher** and **Frank Butler-Jones, B.A.** With an Introduction by **Sir George Beilby, LL.D., F.R.S.** Crown 8vo. 4/- net. ([See p. 48.](#))

The authors show how science has advanced the methods and processes of production and laid the foundation for the establishment of new manufactures. Many important industries are surveyed with the particular object of informing students of chemistry and engineering of the vast field open to them in their professional careers.

The Production and Treatment of Vegetable Oils.

Including chapters on the Refining of Oils, Hydrogenation of Oils, Generation of Hydrogen, Soap Making, Recovery and Refining of Glycerine, and the Splitting of Oils. **T. W. Chalmers, B.Sc., A.M.I.Mech.E.** Imperial 8vo. 9 Folding Plates, 95 Illustrations. 21/- net. ([See p. 52.](#))

Paper Making and its Machinery. **T. W. Chalmers.** Imperial 8vo. 26/- net. Illustrated. ([See p. 52.](#))

Electric Welding and Welding Appliances. **H. Carpmael.** Imperial 8vo. 18/- net. Illustrated. ([See p. 34.](#))

The Gyroscopic Compass. a non-Mathematical Treatment of the Theory and Practical Construction of the Gyroscopic Compass. **T. W. Chalmers.** Demy 8vo. Illustrated. ([See p. 20.](#)) 11/- net.

The “Westminster” Series

Uniform. Extra Crown 8vo. Fully Illustrated.

Soils and Manures. **J. Alan Murray, B.Sc.** 2nd Imp. 8/6 net. ([See p. 59.](#))

The Manufacture of Paper. **R. W. Sindall, F.C.S.** 8/6 net. 2nd Edn. Revised. ([See p. 52.](#))

Timber. **J. R. Baterden, A.M.I.C.E.** 7/6 net.

Electric Lamps. **Maurice Solomon, A.C.G.I., A.M.I.E.E.** 7/6 net.

The Precious Metals: comprising Gold, Silver, and Platinum. **Thomas K. Rose, D.Sc.** 7/6 net.

Decorative Glass Processes. **A. L. Duthie.** 2nd Imp. 8/6 net. ([See p. 53.](#))

The Railway Locomotive. **Vaughan Pendred.** 7/6 net.

Town Gas for Lighting and Heating. **W. H. Y. Webber, C.E.** 7/6 net.

Liquid and Gaseous Fuels, and the Part they Play in Modern Power Production. The late **Vivian B. Lewes, F.I.C., F.C.S.**, Professor of Chemistry, Royal Naval College, Greenwich. 2nd Edn., Edited by **J. B. C. Kershaw, F.I.C.** 12/6 net.

Electric Power and Traction. **F. H. Davies, A.M.I.E.E.** 7/6 net.

Coal. **James Tonge, M.I.M.E.** (Lecturer on Mining at Victoria University, Manchester). 7/6 net.

India-Rubber and its Manufacture. With chapters on Gutta-Percha and Balata. **H. L. Terry, F.I.C., A.Inst.M.M.** 8/6 net.

The Book: Its History and Development. **Cyril Davenport, F.S.A.** 7/6 net.

Glass Manufacture. **Walter Rosenhain**, Superintendent of the Department of Metallurgy in the National Physical Laboratory. 2nd Edn., largely rewritten. 12/6 net. ([See p. 53.](#))

The Law and Commercial Usage of Patents, Designs, and Trade Marks. **Kenneth R. Swan, B.A.(Oxon.)**, of the Inner Temple, Barrister-at-Law. 7/6 net.

Precious Stones. With a chapter on Artificial Stones. **W. Goodchild, M.B., B.Ch.** 7/6 net.

Electro-Metallurgy. **J. B. C. Kershaw, F.I.C.** 7/6 net. ([See p. 39.](#))

Natural Sources of Power. **Robert S. Ball, B.Sc., A.M.I.C.E.** 7/6 net.

Radio-Telegraphy. **C. C. F. Monckton, M.I.E.E.** 7/6 net.

Introduction to the Chemistry and Physics of Building Materials. **Alan E. Munby, M.A.(Cantab.)**. 7/6 net.

The Gas Engine. **W. J. Marshall, A.M.I.M.E.**, and **Captain H. Riall Sankey, R.E., M.I.C.E.** ([See p. 11.](#)) 12/6 net.

Photography. **Alfred Watkins**, Past-President of the Photographic Convention, 1907. 2nd Edn. Revised. 12/6 net. ([See p. 55.](#))

Wood Pulp. **Charles F. Cross, B.Sc., F.I.C.**, **E. J. Bevan, F.I.C.**, and **R. W. Sindall, F.C.S.** 2nd Edn. Revised. 8/6 net. ([See p. 52.](#))

Welding and Cutting of Metals by the Aid of Gases or Electricity. **Dr. L. A. Groth.** 2nd Imp. 8/6 net.

General Foundry Practice. **William Roxburgh.** 3rd Imp. 8/6 net.

Commercial Paints and Painting. **Arthur Seymour Jennings.** 7/6 net.

Textiles and Their Manufacture. New Edition in Preparation.
Demy 8vo. Illustrated. About 18/- net.

A Treatise of Electro-Chemistry.

Edited by **BERTRAM BLOUNT, F.I.C.**, etc.

The Manufacture of Chemicals by Electrolysis. A. J.

Hale. Demy 8vo. 6/- net. ([See p. 39.](#))

Ozone. By **Dr. E. K. Rideal, Ph.D.**, etc. Demy 8vo. 12/- net. ([See p. 39.](#))

Other Volumes in Preparation.

Practical Manuals for Practical Men

Foundations and Machinery Fixing. Francis H. Davies. 2/6
net. ([See p. 28.](#))

Electric Wiremen's Manuals

Testing and Localizing Faults. **J. Wright.** 2/- net.

Arc Lamps and Accessory Apparatus. **J. H. Johnson,**
A.M.I.E.E. 2/- net.

*Motors, Secondary Batteries, Measuring Instruments
and Switchgear.* **S. K. Broadfoot, A.M.I.E.E.** 2nd Edition. 2/-
net.

Mill and Factory Wiring. **R. G. Devey.** 2nd Edition. 2/6 net.

Electric Mining Installations. **P. W. Freudemacher,**
A.M.I.E.E. 2/6 net.

Ship Wiring and Fitting. **T. M. Johnson.** 2/- net.

Bells, Indicators, Telephones, Alarms, etc. **J. B. Redfern**
and **J. Savin.** 2/- net.

*Switchgear and the Control of the Electric Light
Power Circuits.* **A. G. Collis, A.M.I.E.E.** 2/- net.

Thresholds of Science

A SERIES OF HANDY SCIENTIFIC TEXT-BOOKS, WRITTEN IN
SIMPLE, NON-TECHNICAL LANGUAGE, AND ILLUSTRATED
WITH NUMEROUS PICTURES AND DIAGRAMS.

Crown 8vo. 2/6 net per volume.

Mechanics. **C. E. Guillaume.** 200 + xiii pages. 50 Illustrations.

Chemistry. **Georges Darzens.** 122 + x pages. 31 Illustrations.

Botany. **E. Brucker.** 185 + xv pages. 235 Illustrations.

Zoology. **E. Brucker.** 219 + xiii pages. 165 Illustrations.

Mathematics. **C. A. Laisant.** 158 + viii pages. Over 100 Illustrations.

Astronomy. **Camille Flammarion.** 191 + xi pages. 184 Illustrations.

Outlines of Industrial Chemistry

A SERIES OF TEXT-BOOKS INTRODUCTORY TO THE CHEMISTRY
OF THE NATIONAL INDUSTRIES.

Edited by **GUY D. BENGOUGH, M.A., D.Sc.**

An Introduction to the Study of Fuel. **F. J. Brislee, D.Sc.**

With many Illustrations. New Edn. 9/6 net. ([See p. 14.](#))

The Chemistry of Dyeing and Bleaching of Vegetable

Fibrous Materials. **Julius Hübner, M.Sc. Tech., F.I.C.** Demy

8vo. Illustrated. 20/- net. ([See p. 44.](#))

The Chemistry of the Rubber Industry. **Harold E. Potts,**

M.Sc. Demy 8vo. 7/6 net. ([See p. 43.](#))

Iron and Steel. An Introductory Text-Book for Engineers and

Metallurgists. **O. F. Hudson, M.Sc., A.R.C.S.,** and **Guy D. Bengough,**

M.A., D.Sc. Demy 8vo. Illustrated. 8/- net. ([See p. 18.](#))

The Chemistry of the Oil Industries. **J. E. Southcombe,**

M.Sc., F.C.S. Demy 8vo. Illustrated. 9/- net. ([See p. 43.](#))

Cement, Concrete and Bricks. **A. B. Searle.** Demy 8vo.

Illustrated. 10/6 net. ([See p. 43.](#))

Other Volumes in Preparation.

Metallurgy: A Series

Edited by **WALTER ROSENHAIN, F.R.S., D.Sc.**, etc.

An Introduction to the Study of Physical Metallurgy.

Walter Rosenhain, B.A., D.Sc., F.R.S. 12/6 net. ([See p. 17.](#))

Volumes on THE METALLURGY OF BRASS, ZINC, ALUMINIUM, on STRAIN, and the REFRACTORY METALS are in preparation.

The D.-S. Series of Technical Dictionaries in Six Languages

English. Spanish. German. Russian. French. Italian.

With each word, term, or item, so far as it is possible, a diagram, formula, or symbol is given, so that error or inaccuracy is almost impossible.

- Vol. I. THE ELEMENTS OF MACHINERY and the Tools most frequently used in Working Metal and Wood. 10/6 net.
- Vol. II. ELECTRICAL ENGINEERING, including Telegraphy and Telephony. 30/- net.
- Vol. III. STEAM BOILERS AND STEAM ENGINES. 18/- net.
- Vol. IV. INTERNAL COMBUSTION ENGINES. 12/- net.
- Vol. V. RAILWAY CONSTRUCTION AND OPERATION. 15/- net.
- Vol. VI. RAILWAY ROLLING STOCK. 14/- net.
- Vol. VII. HOISTING AND CONVEYING MACHINERY. 12/- net.
- Vol. VIII. REINFORCED CONCRETE IN SUB- AND SUPERSTRUCTURE. 8/6 net.
- Vol. IX. MACHINE TOOLS
(Metal Working; Wood Working). 14/- net.
- Vol. X. MOTOR VEHICLES
(Motor Cars, Motor Boats, Motor Air Ships, Flying Machines). 18/- net.
- Vol. XI. METALLURGY. 12/6 net.
- Vol. XII. PNEUMATICS
(Compressed Air, Air Pumps, Windmills, etc.) 30/- net.
- Vol. XIII. STRUCTURAL ENGINEERING. 22/- net.

“The best book, or series of books, of its kind yet attempted.”—*The Engineer*.

“Anyone desiring a Technical Dictionary for a foreign language will find this the most comprehensive and explicit that has ever been issued.”—*Mechanical Engineer*.

Internal Combustion Engines

Hot Bulb Oil Engines and Suitable Vessels. **Walter Pollock, M.I.N.A., M.I.Mech.E., M.I.M.** Demy 8vo. Fully Illustrated. 42/- net.

CONTENTS.—Scope of Book. Introductory Remarks. Comparison for Power. Demand for Engines. Owner's Requirements. Makes of Engines. Design of Engine. Materials. Workmanship. Hot Bulbs. Fuel Pumps. Oil v. Coal. Advantages, Disadvantages. Supply of Oil Fuels. Designs of Vessels. Coefficients of Fineness. Notes on Working. Stopping the Engine. Attention while Running. Manœuvring and Running Light. Maintenance and Overhauling. Engineers' Log. Tools. Spares for Engines. Superintendence. Notes on Installing. Classification. Lloyd's Scantlings. Port Authorities. Passenger Certificates. Auxiliary Machinery.

The Internal Combustion Engine. Being a Text-Book on Gas, Oil, and Petrol Engines, for the use of Students and Engineers. **H. E. Wimperis, M.A., M.I.E.E.** 3rd Edition Revised and Enlarged. Demy 8vo. 8/6 net.

CONTENTS.—Preface. List of Symbols. Tables of Constants. Molecular Weights of Gases. Elementary. Thermodynamic Cycles, Combustion and Explosion. Thermodynamics. The Gas Engine. The Gas Producer. Blast-Furnace and Coke-Oven Gases. Oil and Petrol Engines. Petrol Engine Efficiency and Rating. Problems and Solutions. Index.

A Primer of the Internal Combustion Engine. A Text-Book for First and Second Year Students. **H. E. Wimperis, M.A., M.I.E.E.,** etc. Crown 8vo. 3rd Imp. 4/- net.

CONTENTS.—List of Symbols. Useful Constants. Introductory. Theory of Heat. Behaviour of Gases and Vapours. The Ideal Engine. The Real Engine. Fuels and Gas Producers. Engine Details. Engine Tests. Bibliography. Examples and Answers. Index.

Governors and the Governing of Prime Movers.
Trinks. ([See p. 16.](#))

Diesel Engine Design. H. F. P. Purday. Demy 8vo. 271 Illustrations. 21/- net.

CONTENTS.—First Principles. Thermal Efficiency. Exhaust, Suction and Scavenge. The Principle of Similitude. Crank. Shafts. Fly-Wheels. Framework. Cylinders and Covers. Running Gear. Fuel Oil System. Air and Exhaust System. Compressed Air System. Valve Gear. Index.

The Diesel Engine for Land and Marine Purposes. 4th Edn. Revised and brought up to date. **A. P. Chalkley, B.Sc., A.M.Inst.C.E., A.I.E.E.** Demy 8vo. 12/6 net.

CONTENTS.—Introduction. General Theory of Heat Engines, with Special Reference to Diesel Engines. Action and Working of the Diesel Engine. Construction of the Diesel Engine. Installing and Running Diesel Engines. Testing Diesel Engines. Diesel Engines for Marine Work. Construction of the Diesel Marine Engine. The Future of the Diesel Engine. Appendix. Index.

High Power Gas Engines. **H. Dubbel.** Translated, edited and expanded to include British Engines and Practice, by **F. Weinreb.** 197 + xii pages. 436 Illustrations. 18/- net.

CONTENTS.—Author's Preface. Translator's Preface. The Cycle of the Gas Engine. Output and Cylinder Dimensions. The Governing of the Four-Cycle Engines. Valve Gear of Four-Cycle Engines. The Two-Cycle Engine. Valve Gears. Ignition. The Cylinders. Valves and their Cooling. Pistons and their Cooling. Stuffing Boxes. General Design of Principal Parts. Calculation of the Flywheel Weight. Starting. Piping.

The Gas Turbine. **Norman Davey.** 248 + xiv pages. 100 Illustrations. Demy 8vo. 15/- net.

The Gas Engine. **W. J. Marshall, A.M.I.M.E.,** and **Captain H. Riall Sankey,** late **R.E., M.I.M.E.,** etc. Extra Crown 8vo. 12/6 net.

CONTENTS.—Theory of the Gas Engine. The Otto Cycle. The Two-Stroke Cycle. Water Cooling of Gas Engine Parts. Ignition. Operating Gas Engines. Arrangement of a Gas Engine Installation. The Testing of Gas Engines. Governing. Gas and Gas Producers.

The Energy-Diagram for Gas. **F. W. Burstall, M.A.,** Chance Professor of Mechanical Engineering in the University of Birmingham. With Descriptive Text. 5/- net. The Diagram sold separately, 2/- net.

American Gas-Engineering Practice. **M. Nisbet-Latta,** Member American Gas Institute, M.Am.Soc.M.E. 142 Illustrations. Demy 8vo. 18/- net.

Gas Engine Design. **Charles Edward Lucke, Ph.D.,** Mechanical Engineering Department, Columbia University, New York City. Demy 8vo. 12/6 net.

Gas Engine Construction. By **Henry V. A. Parsell, Jr.,**
M.A.I.E.E., and **Arthur J. Weed, M.E.** 2nd Edn. Demy 8vo.

Gas, Gasoline, and Oil Engines. **Gardner D. Hiscox, M.E.,**
Author of "Mechanical Movements," "Compressed Air," etc. 22nd
Edn, largely rewritten and brought up to date, by **Victor W. Page,**
M.E. Demy 8vo. 18/- net.

A complete practical work defining the elements of internal combustion, engineering, treating exhaustively on the design, construction and practical application of all forms of gas, gasoline, kerosene and crude petroleum oil engines, describing auxiliary systems, the theory and management of explosive motors for stationary and marine work, automobiles, aeroplanes and motor cycles, also producer gas and its production.

British Progress in Gas Works Plant and Machinery.
C. E. Brackenbury, C.E., Author of "Modern Methods of Saving
Labour in Gas Works." Super-Royal 8vo. 6/- net.

Steam Engines, Boilers, etc.

Governors and the Governing of Prime Movers.

Trinks. 22/6 net.

A Manual of the High-Speed Steam Engine. **H. Keay**

Pratt, A.M.I.Mech.E. Demy 8vo. 6/- net.

CONTENTS.—I. History and General Remarks. II. Steam Cylinders and Adjuncts. III. Valves. IV. Pistons. V. Connecting-rods, Piston-rods, Crank-shafts, etc. VI. Baseplates, Flywheels, etc. VII. Governors. VIII. Cast-iron and other Details. IX. Indicator-Cards and Effort Diagrams. X. Testing. XI. Foundations, Pipe Connections, etc. XII. Practical Notes on the Working of High-Speed Engines. XIII. A Description of Various Engines. Index.

The Modern Steam Engine. Theory, Design, Construction, Use.

John Richardson, M.Inst.C.E. With 300 Illustrations. Demy 8vo. 7/6 net.

Boiler Explosions, Collapses and Mishaps. **E. J. Rimmer,**

B.Sc., etc. Demy 8vo. 4/6 net.

The Steam Engine and Turbine. **Robert C. H. Heck, M.E.**

Demy 8vo. 27/6 net.

Test Methods for Steam Power Plants. A reference book for

the use of Power Station Engineers, Superintendents, and Chemists. **E.**

H. Tenney, B.A., M.E. Ex Crown 8vo. 85 Illustrations. 24/- net.

CONTENTS.—Purchase and Testing of Coal. Investigation of the Economy of Combustion. Treating and Testing Water for Boiler-feed Purposes. Evaporating Tests. Testing Prime Movers. Power Plant Lubricants.

Engine Tests and Boiler Efficiencies. **J. Buchetti.** Translated

and Edited from the 3rd Edition by **Alexander Russell, M.I.E.E.,** etc.

Demy 8vo. 10/6 net.

Boiler Draught. **H. Keay Pratt, A.M.I.M.E.** Crown 8vo. 4/6 net.

CONTENTS.—Draught. Calculations relating to Air. Chimneys. Construction. Artificial Draught. Forced Draught. Induced Draught. A comparison. The Application of Mechanical Draught for Land Installations. The Application of Mechanical Draught in Marine Practice. The Chemistry of Combustion. Index.

Practical Design of Marine Single and Double-Ended Boilers, including Uptakes, etc. **John Gray**, Member of North-East Coast Inst. of Engineers and Shipbuilders. Contains New Board of Trade and Lloyd's rules relating to Marine Boilers. 2nd Edition Revised and Enlarged. Demy 8vo. 12/6 net.

Steam Boilers: Their History and Development. **H. H. Powles, M.I.M.E.**, etc. Imperial 8vo. 25/- net.

Steam Pipes: Their Design and Construction. **W. H. Booth**, Author of "Liquid Fuel," etc.

New Edition in Preparation.

The New Steam Tables. Calculated from Professor Callendar's Researches. **Professor C. A. M. Smith, M.Sc.**, and **A. G. Warren, S.Sc.** 4/6 net.

Experimental Researches on the Flow of Steam through Nozzles and Orifices. To which is added a note on the Flow of Hot Water. **A. Rateau.** Crown 8vo. 5/- net.

Fuel and its Combustion

Fuel, Water and Gas Analysis for Steam Users. **John B. C. Kershaw, F.I.C.** Demy 8vo. 2nd Edn., revised and enlarged. 12/6 net.

CONTENTS.—Origin of Natural Fuels. Physical and Chemical Characteristics of Wood, Peat, Lignite, Bituminous Coal, Semi-Anthracite, Anthracite, and Coke. Liquid Fuels. Briquette Fuels. Sampling and the Preliminary Test for Moisture. Apparatus and Methods used for the Approximate Analysis of Fuels. Special Tests for Sulphur. Apparatus and Methods used for the Calorific Valuation of Solid Fuels. The Practical Application of the Test Results. Forms of Fuel Contract Based upon the Test Results. Sources of Feed-water Supply. Physical and Chemical Characteristics of Well, River, Surface, and Rain Water. Collecting Samples. Apparatus and Methods used for the Approximate Analysis of Feed-waters. Special Tests for Oil. The Practical Applications of the Test Results. The Use of Softening Apparatus and Reagents, and the Tests necessary for obtaining the Best Results. Chemical and Physical Characteristics of the Waste Gases from Various Fuels. Collecting Gas Samples. Apparatus and Methods used for the Approximate Analysis of the Waste Gases. The Applications and Use of Continuous and Recording Gas-testing Apparatus. The Practical Applications of the Gas Test Results. Rules for Sampling Fuel. Typical Tests of English and South Wales Coals. Goutal's Formula for Calculating the Calorific Value from the Results of the Approximate Analysis. The Use of Fuels of Low Calorific Value for Steam-raising Purposes. Tests of Natural and Feed Waters from Various Sources. Typical Tests of Exit Gases.

Powdered Coal as a Fuel. **C. F. Herington.** Medium 8vo. 84 Illustrations. New Edition. 30/- net.

CONTENTS.—Introductory. Coals Suitable for Powdering. Preparation of Powdered Coal. Feeding and Burning. Use in the Cement Industry. Reverberatory Furnaces. In Metallurgical Furnaces. Under Boilers. For Locomotives. Explosions.

Oil Fuel Equipment for Locomotives. **Alfred H. Gibbings, A.M.Inst.C.E.** Demy 8vo. Tables. 8/6 net.

An Introduction to the Study of Fuel. A Text-Book for those entering the Engineering, Chemical and Technical Industries. **F. J. Brislee, D.Sc.** Demy 8vo. 2nd Edn. 9/6 net. (Outlines of Industrial Chemistry.)

CONTENTS.—General Chemical Principles. Weight and Volume of Air required for Combustion. Analysis of Fuel and Flue Gases. Calorimetry and Determination of the Heating Value of a Fuel. Measurement of High Temperatures. Pyrometry. Calculations of Combustion Temperatures. Natural Solid Fuels. Artificial Solid Fuels. Gaseous Fuel. The Manufacture of Producer Gas and Water Gas.

Theory of the Producer Gas and Water Gas Reactions. Explosion and the Explosion Engine. Air Supply and Measurement of Draught. Furnace Efficiency and Fuel Economy. Heat Balances, Furnace and Boiler Tests. Liquid Fuels.

Liquid Fuel and its Combustion. **W. H. Booth, F.G.S., M.Am.Soc.C.E.**, Author of "Water Softening and Treatment," "Steam Pipes: Their Design and Construction." With Illustrations and Diagrams. Imperial 8vo. 24/- net.

Liquid Fuel and its Apparatus. **William H. Booth, F.G.S., M.Am.Soc.C.E.**, Author of "Liquid Fuel and its Combustion," etc. Demy 8vo. New Edition. 9/- net.

Part I. THEORY AND PRINCIPLES. Part II. PRACTICE. Part III. TABLES AND DATA.

Low Grade and Waste Fuels for Power Generation. **J. B. C. Kershaw, F.I.C.** Demy 8vo. Illustrated. 17/- net.

CONTENTS.—Part I. *Fuels*: Introduction. Peat. Lignite, Bagasse and Wood Waste. Coke and Coke-Breeze. Culm and Washery. Waste. Towns' Refuse and Garbage. Pitch. Waste Gases.—Part II. *Scientific Control*: Fuel Sampling and Testing. Calorimetric Tests. Boilers and Furnaces. Automatic CO₂ Apparatus. Index.

Liquid and Gaseous Fuels and the Part they Play in Modern Power Production. **Vivian B. Lewes, F.I.C., F.C.S.**, late Professor of Chemistry, Royal Naval College, Greenwich. 2nd Edition Revised and Enlarged. Edited by **J. B. G. Kershaw, F.I.C.** Ex. Crown 8vo.

Coal. **James Tonge, M.I.M.E., F.G.S.**, etc. Lecturer on Mining at Victoria University, Manchester. Ex. Crown 8vo. 7/6 net.

Domestic Fuel Consumption. **A. H. Barker, B.A., B.Sc.**, Author of "The Theory and Practice of Heating and Ventilation," etc. Demy 8vo. 180 pp. 14/- net.

Written so as to be readily comprehensible by any person without scientific training, this book will be read with interest and profit by all who, whether for private or public purposes, are desirous of checking the present lavish waste of our fuel resources.

Oil Firing for Kitchen Ranges and Steam Boilers. **E. C. Bowden-Smith, M.I.Mech.E.** Demy 8vo. 9/- net.

A concise and practical account of the methods of employing heavy residues and fuel oil in a kitchen range for culinary purposes, and also for steam boilers.

Machinery, Power Plants, etc.

Theory and Practice of Mechanics. **S. E. Slocum, B.E., Ph.D.** Demy 8vo. 442 + xlii pages. Illustrated with diagrams, etc. 15/- net.

Dynamometers. **Rev. Frederick John Jervis-Smith, M.A., F.R.S.** Edited and amplified by **Charles Vernon Boys, F.R.S.** 118 Illustrations. Demy 8vo. 14/- net.

CONTENTS.—Introduction. Friction. Planimeters. Friction Brakes. Water Brakes. Air Brake. Magnetic Brake Dynamometer. End Thrust Brakes. Historical. Transmission. Dynamometers. Torsion Power Measuring Machines. Torsion Power-Measuring Machines by different Inventors. The Cradle Dynamometer. Dynamometric Tests of Motor-Car Engines. Model Ship Dynamometers used in Aeronautics.

A Handbook of Testing. **C. A. Middleton Smith, B.Sc., A.M.I.M.E., A.M.I.E.E.** Professor of Engineering in the University of Hong Kong, late of East London College; Author of “Lectures on Suction Gas Plants,” etc. I. MATERIALS. 2nd Edn. 14/- net. II. PRIME MOVERS. In Preparation.

Mechanical Movements, Powers, Devices and Appliances. **Gardner D. Hiscox, M.E.,** Author of “Gas, Gasoline, and Oil Engines,” etc. Over 400 pages. 1646 Illustrations and Descriptive Text. Demy 8vo. 18/- net.

Mechanical Appliances. Supplementary Volume to Mechanical Movements. **Gardner D. Hiscox, M.E.** 400 pages. About 1000 Illustrations. Demy 8vo. 18/- net.

Mechanical Technology. **G. F. Charnock, M.Inst.C.E., M.Inst.M.E.** Demy 8vo. 650 pages. 12/6 net.

This work, designed chiefly for engineering students, deals in a comprehensive manner with the preparation and properties of the raw material, the production of castings by taking advantage of the

property of fusibility, and the production of bars, plates, forgings, etc., by utilising the properties of malleability and ductility. It is perhaps the most complete manual on the subject in any language.

The Elements of Mechanics of Materials. A Text-Book for Students in Engineering Courses. **C. E. Houghton, A.B., M.M.E.** Demy 8vo. 10/6 net.

Precision Grinding. A Practical Book on the Use of Grinding Machinery for Machine Men. **H. Darbyshire.** Pages viii + 162. Demy 8vo. 6/6 net.

Forging of Iron and Steel. A Text-Book for the use of Students in Colleges, Secondary Schools and the Workshop. **W. Allyn Richards.** Demy 8vo. 337 Illustrations. Price 7/6 net.

CONTENTS.—Historic Use of Iron and Steel. Iron and Steel. Equipment. Fuel and Fires. Drawing Down and Upsetting. Bending and Twisting. Splitting, Punching, and Riveting. The Uses of Blacksmiths' Tools. Welding. Electric, Autogenous and Thermit Welding. Brazing. Tool Steel. High Speed Tool Steel. Art Ironwork. Steam and Power Hammers. Calculations. Appendix. Index.

Water Softening and Treatment: Condensing Plant, Feed Pumps and Heaters for Steam Users and Manufacturers. **W. H. Booth, F.G.S., M.Am.Soc.C.E.** 2nd Edn. With Tables and many Illustrations. Demy 8vo. 9/6 net.

Dock and Lock Machinery. **William Hunter, M.Inst.C.E.** Demy 8vo. Illustrated.

Dies: Their Construction and Use for the Modern Working of Sheet Metals. **Joseph V. Woodworth.** 384 pages. Demy 8vo. 31/6 net.

Governors and the Governing of Prime Movers. **W. Trinks, M.E.** Demy 8vo. 22/6 net.

Shop Kinks. A book for Engineers and Machinists, showing Special Ways of doing Better, Cheap, and Rapid Work. **Robert Grimshaw.** 4th Edn. With 222 Illustrations. Demy 8vo. 24/- net.

Cams, and the Principles of their Construction. **George Jepson**, Instructor in Mechanical Drawing in the Massachusetts Normal School. Demy 8vo. 12/- net.

The Economic and Commercial Theory of Heat Power-Plants. **Robert H. Smith**, A.M.Inst.C.E., M.I.M.E., M.I.E.E., etc. Prof. Em. of Engineering and Mem. Ord. Meiji. Numerous Diagrams. Royal 8vo. 24/- net.

Entropy: Or, Thermodynamics from an Engineer's Standpoint, and the Reversibility of Thermodynamics. **James Swinburne**, M.Inst.C.E., M.I.E.E., etc. Illustrated with Diagrams. Crown 8vo. 5/- net.

Compressed Air: Its Production, Uses and Applications. **Gardner D. Hiscox**, M.E., Author of "Mechanical Movements Powers, Devices," etc. 5th Edn. Revised and Enlarged. 665 pages. 540 Illustrations and 44 Air Tables. Demy 8vo. 25/- net.

Natural Sources of Power. **Robert S. Ball**, B.Sc., A.M.Inst.C.E. With 104 Diagrams and Illustrations. Ex. Crown 8vo. 7/6 net.

Pumps

The Efficiency of Pumps and Ejectors. **E. C. Bowden-Smith, M.I.Mech.E.,** etc. Late Egyptian Civil Service. Demy 8vo. Illustrated. 14/6 net.

Centrifugal Pumps. **Louis C. Loewenstein, E.E., Ph.D.,** and **Clarence P. Crissey, M.E.** 320 Illustrations. 26/- net.

CONTENTS.—Theory of Centrifugal Pumps. Consumption of Power and Efficiency. The Calculation of Impeller and Guide Vanes. Design of Important Pump Parts. Types of Centrifugal Pumps. Testing of Centrifugal Pumps.

Iron, Steel, and other Metals

An Introduction to the Study of Physical Metallurgy.

Walter Rosenhain, B.A., D.Sc., F.R.S., Superintendent, Metallurgy Department of the National Physical Laboratory. With 32 Plates and 131 Illustrations in the Text. Demy 8vo. 12/6 net. ([See p. 49.](#))

The Basic Open-Hearth Steel Process. **Carl Dichmann.**

Translated and Edited by Alleyne Reynolds. 2nd Imp. Demy 8vo. Numerous Tables and Formulæ. 12/6 net.

Iron and Steel. An Introductory Text-Book for Engineers and Metallurgists. (Outlines of Industrial Chemistry.) **O. F. Hudson, M.Sc., A.R.C.S.**, Lecturer on Metallography, Birmingham University, with a Section on Corrosion by **Guy D. Bengough, M.A., D.Sc.**, Lecturer in Metallurgy, Liverpool University; Investigator to the Corrosion Committee of the Institute of Metals. Demy 8vo. 8/- net.

Forging of Iron and Steel. **W. Allyn Richards.** Demy 8vo. 337 Illustrations. 7/6 net.

CONTENTS.—Historic Use of Iron and Steel. Iron and Steel. Equipment. Fuel and Fires. Drawing Down and Upsetting. Bending and Twisting. Splitting, Punching and Riveting. The Uses of Blacksmiths' Tools. Welding. Electric, Autogenous and Thermit Welding. Brazing. Tool Steel. High Speed Tool Steel. Art Ironwork. Steam and Power Hammers. Calculations. Appendix. Index.

Electro-Thermal Methods of Iron and Steel

Construction. **J. B. C. Kershaw, F.I.C.** With an Introduction by **Dr. J. A. Fleming, F.R.S.** 239 + xxiii pages. 50 Tables and 92 Diagrams and Photographs. Demy 8vo. 9/- net.

Malleable Cast Iron. **S. Jones Parsons, M.E.** New Edn. 14/- net.

CONTENTS.—Melting. Moulding. Annealing. Cleaning and Straightening. Design. Patterns. Inspection and Testing. Supplementary Processes. Application.

General Foundry Practice. Being a Treatise on General Iron Founding, Job Loam Practice, Moulding and Casting of Finer Metals, Practical Metallurgy in the Foundry, etc. **William Roxburgh, M.R.S.A.** With over 160 Figures and Illustrations. Extra Crown 8vo. 8/6 net.

Hardening, Tempering, Annealing and Forging of Steel. A Treatise on the Practical Treatment and Working of High and Low Grade Steel. **Joseph V. Woodworth.** With 201 Illustrations. Demy 8vo. 5th Edition. 21/- net.

Welding and Cutting of Metals by the aid of Gases or Electricity. **Dr. L. A. Groth.** Ex. Crown 8vo. 8/6 net.

The Precious Metals: Comprising Gold, Silver, and Platinum. **Thomas K. Rose, D.Sc.** Ex. Crown 8vo. 7/6 net.

Motor-Cars and Engines

Text-Book of Motor-Car Engineering. **A. Graham Clarke,**
M.I.A.E., A.M.I.M.E.

Vol. I. CONSTRUCTION. Demy 8vo. 15/- net.

CONTENTS.—The General Principles and Construction of the Petrol Engine. Details of Engine Construction. Petrol. Fuels other than Petrol. Carburetters and Carburation. Thermodynamics of the Petrol Engine. Horse-power. Mechanical (Thermal and Combustion Efficiencies). The Principles and Construction of Coil and Accumulator Ignition. Magneto Ignition. Engine Control Systems. Engine Cooling Systems. Crank Effort Diagrams, Clutches and Brakes. Change-Speed Gears. Transmission Gear. Steering Gears. Lubricants. Lubrication. Ball and Roller Bearings. Chassis Construction. General Principles of the Steam Car. Steam Engines and Condensers. Steam Generators and Pipe Diagrams. The Electric Car. Materials used in Motor-Car Construction. Syllabus of City and Guilds of London Institute in Motor-Car Engineering. Examination Papers. Physical Properties of Petrols. Mathematical Tables and Constants.

Vol. II. DESIGN. 16/- net.

CONTENTS.—Introduction. Materials of Construction. General Considerations in Engine Design. Power Requirements. Determination of Engine Dimensions. Cylinders and Valves. Valve Gears. Pistons, Gudgeons, and Connecting Rods. Crank-shafts and Fly-wheels. The Balancing of Engines. Crank-cases and Gear-boxes. Engine. Lubrication and Cooling Arrangements. Inlet, Exhaust and Fuel Piping. Clutches and Brakes. Gearing. Transmission Gear. Frames, Axles and Springs. Torque and Radius Rods. Steering Gears.

Henslowe's Motor Dictionary. Compiled by **Leonard P. Henslowe.** English-French, French-English. Tables of British and Metric Units of Measurement.

Aeronautics

Air screws. **A. Fage, A.R.C.Sc., D.I.C.,** etc. Aerodynamics Department, The National Physical Laboratory. Cr. 4to. Fully illustrated. 34/- net.

A Primer of Air Navigation. **H. E. Wimperis, M.A.** Author of "The Internal Combustion Engine." Cr. 8vo. Fully Illustrated. 8/6 net.

The Resistance of the Air and Aviation. Experiments conducted at the Champ-de-Mars Laboratory, by **G. Eiffel.** 2nd Edn. Revised and Enlarged. Translated by **Jerome C. Hunsaker,** Assistant Naval Constructor United States Navy. Over 130 Illustrations and Diagrams. Royal 4to. 42/- net.

Dynamics of Mechanical Flight. **Sir George Greenhill,** late Professor of Mathematics in the R.M.A., Woolwich. Demy 8vo. 6/- net.

Aircraft in Warfare: The Dawn of the Fourth Arm. **F. W. Lanchester, M.Inst.C.E., M.Inst.A.E.,** Member Advisory Committee for Aeronautics. With Introductory Preface by **Maj.-Gen. Sir David Henderson, K.C.B.** Medium 8vo. 12/6 net.

Aerial Flight. **F. W. Lanchester.**

Vol. I. AERODYNAMICS. Demy 8vo. 21/- net. 4th Edn. Revised. 460 pages. 160 Illustrations.

CONTENTS.—Fluid Resistance and Its Associated Phenomena. Viscosity and Skin Friction. The Hydronamics of Analytical Theory. Wing Form and Motion in the Periptery. The Aeroplane. The Normal Plane. The Inclined Aeroplane. The Economics of Flight. The Aerofoil. On Propulsion, the Screw Propeller, and the Power Expended in Flight. Experimental Aerodynamics.

Vol. II. AERODONETICS. Demy 8vo. 21/- net. 2nd Edition. Revised. 450 pages. 208 Illustrations.

CONTENTS.—Free Flight, General Principles and Phenomena. The Phugoid Theory. The Equations of the Flight Path. The Flight Path Plotted. Elementary Deductions from the Phugoid Theory.

Stability of the Flight Path as affected by Resistance and Moment of Inertia. Experimental Evidence and Verification of the Phugoid Theory. Lateral and Directional Stability. General Conclusions. Soaring. Experimental Aerodnetics. Appendices on the Theory and Application of the Gyroscope. The Flight of Projectiles, etc.

The Flying Machine from an Engineering Point of View. Together with a Discussion concerning the Theory of Sustentation and Expenditure of Power in Flight. **F. W. Lanchester, M.Inst.C.E.** Demy 8vo. 5/- net.

Stability and Equilibrium of Floating Bodies. **Bernard C. Laws, B.Sc., A.R.C.Sc., A.M.Inst.C.E., M.I.N.A.** 130 Illustrations. Demy 8vo. 10/6 net.

Marine Engineering.

The Gyroscopic Compass. (“The Engineer” Library.) **T. W. Chalmers, B.Sc.,** etc. Demy 8vo. Fully illustrated. 11/- net.

The Design of Marine Engines and Auxiliaries. 110 Illustrations and Folding Plans. **Edward M. Bragg, S.B.** Medium 8vo. 12/6 net.

CONTENTS.—Determination of Cylinder Dimensions. Design of Engine Parts. Engine Balancing. Condensers and Air Pumps. Turning Engines and Reversing Engines.

Hot Bulb Oil Engines and Suitable Vessels. **Walter Pollock.** Demy 8vo. 42/- net. ([See p. 10.](#))

Stability and Equilibrium of Floating Bodies. **Bernard C. Laws, B.Sc., A.R.C.Sc., A.M.Inst.C.E., M.I.N.A.** 130 Illustrations. Demy 8vo. 10/6 net.

CONTENTS.—General Considerations. Ships. Amsler’s Integrator, Submarines. Floating Docks. Aircraft. Caissons. Index.

Practical Design of Marine Single and Double-Ended Boilers. **John Gray** ([See p. 12.](#))

Ship Form, Resistance, and Screw Propulsion. **G. S. Baker, M.Inst.Naval Architects.** Medium 8vo. 2nd Edn. Treated from the practical point of view for the use of naval architects, engineers, and draughtsmen. 16/-net.

CONTENTS.—Nomenclature. Steam Line Motion. Skin Friction Resistance. Eddy-making. Waves and Wave-making. Ship Model Experiments. Dimensions and Form. Curve of Areas. Shape and Fineness of Ends, with Parallel Body. Position of Maximum Section, and Relative Length of Entrance and Run. Midship Section Area and Shape. Level Lines and Body Plan Sections. Racing and other High Speed Vessels. Appendages. Restricted Water Channels. Screw Propeller Nomenclature and Geometry. Theories of the Screw Propeller Nomenclature and Geometry. Theories of the Screw Propeller. The Elements of Propulsion. Screw Propellers in Open Water. Propeller Blades. Hull Efficiency, Wake and Thrust Deduction. Main Engines. Cavitation. Measured Mile Trials.

The Shipbuilding Industry. **Roy W. Kelly and Frederick J. Allen.** Preface by **Charles M. Schwab.** Illustrated. Crown 8vo. 21/- net.

Cold Storage, Heating and Ventilating on Board Ship. **Sydney F. Walker, R.N.** Extra Crown 8vo. 18/- net.

The Elements of Graphic Statics and of General Graphic Methods. **W. L. Cathcart, M.Am.Soc.M.E., and J. Irvin Chaffee, A.M.** 159 Diagrams. 21/6 net.

This book is designed for students of marine and mechanical engineering and naval architecture. It reviews the principles of graphics and their application both to frame structures and to mechanism.

Handbook for the Care and Operation of Naval Machinery. **Commander H. C. Dinger,** United States Navy. ([See p. 57.](#)) 3rd Edition. 21/- net.

Turbines Applied to Marine Propulsion. **Stanley J. Reed, A.M.Inst.C.E.** Crown 4to. Over 100 Illustrations and Diagrams. 2nd Edn. Revised. 16/- net.

CONTENTS.—General Properties of Steam—Two-Stage Rateau Turbine Nozzles. The General Design and Arrangement of Turbines. The Design and Construction of Parsons Turbine. The Design and Construction of Curtis Turbine. Combined or Mixed Turbines. Steam Thrust of Reaction Turbines. Superheated Steam. Cavitation. Relative Effect of High Vacua with Turbines and Reciprocating Engines. Utilisation of Auxiliary Engine Exhaust in the Turbines. Astern Turbines.

Hydraulics and Turbines.

Ship Form, Resistance, and Screw Propulsion.

([See above.](#))

Stability and Equilibrium of Floating Bodies. **Bernard C. Laws, B.Sc., A.R.C.Sc., A.M.Inst.C.E., M.I.N.A.** ([See p. 20.](#))

Hydraulics and its Applications. A Text-Book for Engineers and Students. **A. H. Gibson, D.Sc., M.Inst.C.E.** 2nd Edn. Revised and Enlarged. Demy 8vo. 24/- net.

“One of the most satisfactory text-books on Hydraulics extant.”—*Mechanical World.*

Hydraulics, Text-book of. Including an Outline of the Theory of Turbines. **L. M. Hoskins**, Professor of Applied Mathematics in the Leland Stanford Junior University. Numerous Tables. Demy 8vo. 16/- net.

Hydraulics of Pipe Lines. **W. F. Durand**, Professor of Mechanical Engineering, Leland Stanford University, California. Demy 8vo. Illustrated.

Water Hammer in Hydraulic Pipe Lines. Being a Theoretical and Experimental Investigation of the Rise or Fall of Pressure in a Pipe Line, caused by the gradual or sudden closing or opening of a Valve; with a Chapter on the Speed Regulation of Hydraulic Turbines. **A. H. Gibson, D.Sc., M.Inst.C.E.** 2nd Edn. Crown 8vo. 8/- net.

Conveyance and Distribution of Water for Water Supply. Aqueducts, Pipe Lines and Distributing Systems. **Edward Wegmann, C.E.** A Practical Treatise for Water-Works Engineers and Superintendents. Medium 8vo. 367 Illustrations. 8 Plates. 663 pages. 32/6 net.

CONTENTS.—Part I: Water Consumption and Hydraulic Formulas. Consumption of Water. Flow of Water in Aqueducts. Flow of Water through Orifices and Pipes. Part II: Design and Construction. Wooden Pipes. Cast Iron Pipes. Wrought Iron and Steel Pipes. Pipes of Vitrified Stone Ware, Cement, and Concrete. Stresses in Water Pipes. Flexible Pipe. Joints. Submerged Pipes. Gates and Valves. Hydrants. Intake Pipes and Tunnels. Aqueducts. Description of Aqueducts. Service Reservoirs. Stand Pipes. Cylindrical Tanks of Wood or Steel. Fire Protection. High Pressure Water Systems. Distribution Systems. Part III: Maintenance and Operation. Service Pipes and Connections. Cleaning Aqueducts and Water Mains. Thawing Frozen Pipes and Hydrants. Leakage from Aqueducts, Mains, and Service Pipes. Durability of Wooden and Metal Pipes. Electrolysis. Tools and Machines for Pipe Work. Detection and Prevention of Waste of Water. Pitot Tube Gauging. Water Meters. Recording Instruments. Appendix I: Standard Specifications of American Water Works Association for “Cast Iron Water Pipe and Special Castings.” Tables of Wrought Iron, Steel, and *Universal* Cast Iron Pipes. Appendix II: Standard Specifications of American Water Works Association for Hydrant and Valves. Appendix III: Standard Specifications of Association of American Steel Manufacturers for “Structural and Boiler Steel.” Appendix IV: Special Fire Protection for Buildings. Appendix V: Fire Steam Tables prepared by National Board of Fire Underwriters. Index.

Hydraulic Tables for the Flow of Water in Circular Pipes under Pressure, Timber Flumes, Open Channels and Egg-shaped Conduits. **Joseph H. Harper, C.E.** Crown 8vo. 192 pages. With 22 Charts. 8/6 net.

Water Pipe and Sewage Discharge Diagrams. **T. C. Ekin, M.Inst.C.E., M.I.M.E.** ([See p. 31.](#))

Hydroelectric Developments and Engineering. A Practical and Theoretical Treatise on the Development, Design, Construction, Equipment and Operation of Hydroelectric Transmission Plants. **Frank Köester.** 42/- net.

Modern Turbine Practice and Water Power Plants. With Terms and Symbols used in Hydraulic Power Engineering. **John Wolf Thurso,** Civil and Hydraulic Engineer. Royal 8vo. 21/- net.

Mining.

Electric Mine Signalling Installations. A Practical Treatise on the Fitting-up and Maintenance of Electrical Signalling Apparatus in Mines. **G. W. Lummis Paterson.** Crown 8vo. 140 Illustrations. 5/- net. ([See p. 34.](#))

Physics and Chemistry of Mining and Mine Ventilation. A Practical Handbook. **Joseph J. Walsh.** Extra Crown 8vo. Second Revised and Enlarged Edition. 12/6 net.

CONTENTS.—Matter. Motion, Velocity and Force. Gravitation. Liquids and Liquid Pressure. Heat. Gases. The Barometer. Specific Heat. Air Analysis. Mine Ventilation. Formulas. Mine Fires.

Application of Electric Power to Mines and Heavy Industries. **W. H. Patchell, M.Inst.C.E., M.I.M.E., M.Am.I.E.E.** Demy 8vo. 12/6 net.

CONTENTS.—Electricity in Mines. Cables. Coal Cutters and Underground Details. Haulage Gears. Rating of Haulages. Winding Engines. Types of Electric Winders. Ventilation and Air Compressing. Pumping. Rolling Mills. Machine Tools and Cranes, Electric Welding and Furnaces.

Railway Engineering.

Railway Maintenance Engineering. With Notes on Construction. **William H. Sellew, A.S.M.E.**, Author of "Steel Rails, their History, Properties, Strength and Manufacture," Member of the American Railway Engineering Association. Extra Crown 8vo. 194 Illustrations. 19/- net.

CONTENTS.—Engineering. Land. Grading. Bridges, Trestles and Culverts. Ties. Rails. Other Track Material. Ballast. Maintaining Track and Right of Way. Station and Roadway Buildings. Water Stations. Fuel Stations. Shops and Engine Houses. Icing Stations. Signals and Interlockers.

Locomotive Valves and Valve Gears. **J. H. Yoder and G. B. Wharen.** Fully Illustrated. Large 8vo. 14/- net.

CONTENTS.—Common D Slide Valve. Valve Events. Description of Slide Valve and Its Seat. Special Designs of Slide Valves. Balancing the Slide Valve. The Piston Valve. Piston Valve for Balanced Compound Locomotives. The Stephenson Valve Gear. Effects of Raising or Lowering the Link. The Walschaert Valve Gear. Construction and Development. Description of Parts. Operating Parts of the Gear. Relative Crank Pin and Piston Positions for Different Valve Events at Full Forward Gear. Relative Crank Pin and Positions for Different Valve Events at the Running Cut-off Position. Distortions of the Walschaert Valve Gear. Layout of Walschaert Gear. Baker Locomotive Valve Gear. The Young Locomotive Valve, Valve Gear and Reverse Gear. The Gooch Stationary Link. The Allen Valve Gear. Effects of Altering the Valve and Its Events. Locomotive Valve Setting. Stephenson Gear. Walschaert Gear. Rules for Valve Setting. Baker Locomotive Valve Gear. Summary. The Indicator Diagram and its Application. Application of the Diagram. Horse Power of Locomotives. Tractive Force of Two-cylinder Compound Locomotives. Tractive Force of Mallet Articulated Compound Locomotives. Classification of Locomotives.

Oil Fuel Equipment for Locomotives. **Alfred H. Gibbings, A.M.Inst.C.E.** Demy 8vo. Tables. 8/6 net.

Steel Rails: Their History, Properties, Strength and Manufacture. With Notes on the Principles of Rolling Stock and Track Design. **William H. Sellew.** 361 Illustrations. 33 Folding Plates. 576 pages. 52/- net.

CONTENTS.—Development of the Present Section. Pressure of the Wheel on the Rail. Supports of the Rail. Stresses in the Rail. Strength of the Rail. Influence of Detail of Manufacture. Rail Specifications. Appendix.

Solution of Railroad Problems by the Slide Rule. **E. R. Cary, M.Am.S.C.E.** 43 Diagrams. 4/6 net.

The Practical Railway Spiral. With short Working Formulas and Full Tables of Deflection Angles. Complete Notes of Illustrative Examples. **L. C. Jordan, B.S.** 6/- net.

The Transition Spiral. **A. L. Higgins, A.R.C.S.** Crown 8vo. Illustrated. About 7/6 net.

Single-Phase Electric Railways. **E. Austin.** Demy 4to. 346 Illustrations. 21/- net. ([See p. 33.](#))

Reinforced Concrete Railway Structures. **J. D. W. Ball, A.M.Inst.C.E.** Demy 8vo. 9/- net. (Glasgow Text-Books of Civil Engineering.)

CONTENTS.—Preliminary Considerations. Bending Stresses. Shear Stress. Floors and Buildings. Foundations and Rafts. Retaining Walls. Bridges. Arched Bridges. Sleepers, Fence Posts, etc. Summary of Notation.

The Design of Simple Steel Bridges. **P. O. G. Usborne, late R.E.** 12/6 net.

CONTENTS.—Definitions. Bending Moments. Moments of Flexure. Moments of Resistance. Shear Deflection. Solid Beams and Examples. Struts and Ties. Rivets and Joints. Rolling Loads. Bending Moments, Shear Stresses. Loads. Plate Girders. Bridge Floors. Railways, Roads. Principles of Bridge Design. Plate Girders. Parallel Braced Girders. Braced Girders (2). Shop Practice and General Details.

Reinforced Concrete Bridges. **Frederick Rings, M.S.A., M.C.I.** ([See p. 26.](#))

Railway Signal Engineering (Mechanical). **Leonard P. Lewis**, of the Caledonian Railway, Glasgow. Illustrated. Demy 8vo. 2nd Edition. (Glasgow Text-Books of Civil Engineering.) 14/- net.

CONTENTS.—Introduction. Board of Trade Rules. Classes and Uses of Signals. Constructional Details of Signals. Point Connections. Interlocking Apparatus. Signal Box Arrangements. Miscellaneous Apparatus. Signalling Schemes. Interlocking Tables, Diagrams, etc. Methods of Working Trains. Appendix.

Railway Terms. Spanish-English, English-Spanish, Dictionary of. **André Garcia.** 12/6 net.

The Railway Locomotive: What it is, and Why it is What it is. **Vaughan Pendred, M.I.M.E., M.I. and S.Inst.** Extra Crown 8vo. 7/6 net.

Earthwork in Railway Engineering. **J. W. P. Gardner, M.I.C.E.** Demy 8vo.

The Field Engineer. **William Findlay Shunk.** 21st Edn. Revised and Enlarged. Foolscap 8vo. Leather. 14/- net.

Reinforced Concrete and Cement.

A Treatise on Cement Specifications. Including the General Use, Purchase, Storage, Inspection and Test Requirements of Portland, Natural, Puzzolan (slag), and Silica (sand) Cement, and Methods of Testing and Analysis of Portland Cement. **Jerome Cochran, B.S., C.E., M.C.E.**, Author of "Inspection of Concrete Construction," etc. Demy 8vo. 100 pages. 6/- net.

Reinforced Concrete Railway Structures. **J. D. W. Ball, A.M.Inst.C.E.** ([See p. 25.](#))

Reinforced Concrete Bridges. **Frederick Rings, M.S.A., M.C.I.** Over 300 Illustrations and Diagrams. Imperial 8vo. 21/- net.

CONTENTS.—List of Symbols. Introductory. Bending Moments, Stresses and Strains. Loads on Bridges and External Stresses. Culverts, Coverings, Tunnels, etc. Beam Bridges. Calculation of Girder Bridges. Examples of Girder Bridges. Design of Arched Bridges and Abutments. Theory of the Arch. Examples of Arched Bridges. Formulæ, Notes, Schedules and other useful information.

Reinforced Concrete Diagrams. **J. Williamson.** Imperial 8vo. 48 pages. 18 Diagrams. 12/- net.

A Manual of Cement Testing. For the use of Engineers and Chemists in Colleges and in the Field. **W. A. Richards, B.S. in M.E.,** and **H. B. North, D.Sc.** 137 + x pages. 54 Diagrams. Demy 8vo. 11/- net.

Manual of Reinforced Concrete. 3rd Edn. Re-written and Enlarged, with the addition of much useful information and many New Tables and Diagrams. **Charles F. Marsh, M.Inst.C.E., M.Am.Soc.E., M.I.Mech.E.** and **William Dunn, F.R.I.B.A.** Pocket size. Limp Cloth. 500 pages. 79 Tables of Data and 191 Diagrams. 14/- net.

A Concise Treatise on Reinforced Concrete. A Companion to "The Reinforced Concrete Manual." **Charles F. Marsh,**

M.Inst.C.E., M.Am.Soc.E., M.I.Mech.E. Demy 8vo. 12/6 net.

Properties and Design of Reinforced Concrete.

Instructions, Authorised Methods of Calculation, Experimental Results and Reports by the French Government Commissions on Reinforced Concrete. **Nathaniel Martin, A.G.T.C., B.Sc., A.M.Inst.C.E.**, former Lecturer on Reinforced Concrete in the Royal West of Scotland Technical College. 8/- net.

Reinforced Concrete Compression Member Diagram.

A Diagram giving the loading size and reinforcements for compression members according to the data and methods of Calculation recommended in the Second Report of the Joint Committee on Reinforced Concrete. **Charles F. Marsh, M.Inst.C.E., M.Am.Soc.E., M.I.Mech.E.** In Cloth covers, mounted on Linen, 5/- net. Unmounted, 3/6 net.

Cement, Concrete and Bricks. **A. B. Searle.** ([See p. 43.](#))

Civil Engineering, Building Construction, etc.

Civil Engineers' Pocket Book. A Reference-Book for Engineers, Contractors and Students, containing Rules, Data, Methods, Formulæ and Tables. **A. I. Frye, S.B., M.Am.S.C.E.** 26/- net.

Some Modern Methods of Ventilation. With special reference to Public Buildings: Standards of Ventilation, Design of Ducts, Selection of Fans, Washers, and Heaters, Specifications, Test Forms, and Specimen Schedules for Designers. **Ronald Grierson, A.M.I.Mech.E., A.M.I.E.E.** Demy 8vo. 8/6 net.

CONTENTS.—The General Principles of Ventilation. The Apparatus Employed Considerations to be observed in the Design of the Air Circuit. Ducts and Registers. Fans. Air Washing and Humidifying Plant (Filters). Heaters. Ozone. Instruments. Guarantees for Complete Installations. Tests. Motor Drives. Suggested Schedules for Designing Complete Installation. Purchase Specifications. The Application of Wet Air Filters to the Cooling of Electrical Machinery.

A Treatise on Cement Specifications. **Jerome Cochran, B.S., C.E., M.C.E.** ([See p. 26.](#))

The Principles of the Application of Power to Road Transport. **H. E. Wimperis, A.M.Inst.C.E., A.M.I.E.E.** Illustrated. Crown 8vo. 5/- net.

The Design of Simple Steel Bridges. **P. O. G. Usborne, late R.E.** ([See p. 25.](#))

The Elements of Structural Design. **Horace R. Thayer.** Medium 8vo.

Vol. I. ELEMENTS. 18/- net.

Vol. II. DESIGN OF SIMPLE STRUCTURES. Discusses the design of simple structures, such as beams, girders, viaducts, trusses, buildings, stand pipes, and elevated tanks. 500 pages. 16/- net.

Reinforced Concrete Railway Structures. **J. D. W. Ball, A.M.Inst.C.E.** ([See p. 25.](#))

Practical Design of Steel-Framed Sheds. **Albert S. Spencer.** Royal 8vo. 12/- net.

CONTENTS.—The Resistance of Steel-Framed Sheds to Wind Forces. The Standardisation of Shed Members and Foundations. Sheeted Sheds. Standard Sections for Sheds with Open Sides and Gables. Sheeted Roofs. Standardising of Shed Members and Foundations for Sheds having Slated Roofs and Brick-panelled Walls. Standard Sections for Sheds with Open Sides and Gables. Slated Roofs. Standard Stresses and Sections for Roof Principals. External Coverings. Gutters, Pipes, Ventilation, Doors, Windows and Skylights.

A Study of the Circular-Arc Bow-Girder. **A. H. Gibson, D.Sc., A.M.Inst.C.E., M.I.Mech.E., and E. G. Ritchie, B.Sc.** 10/6 net.

Masonry Applied to Civil Engineering. A Practical Treatise on the Design and Construction of Engineering Works in Stone and Heavy Concrete. **F. Noel Taylor, M.Inst.C.E.,** Author of “A Manual of Civil Engineering Practice.” 242 pages. Demy 8vo. 7/6 net.

CONTENTS.—Stones used in Constructional Work. General Remarks upon Masonry. Retaining Walls and Earth Pressures. Dock, River, Canal and Sea Walls. Masonry Dams. Masonry Bridges and Arches. Masonry Towers, etc. Monolithic and Block Concrete Construction. Shoring and Underpinning. Index.

Foundations and Machinery Fixing. **Francis H. Davies, A.M.I.E.E.** 2/6 net.

CONTENTS.—The Functions of Foundations, Nature of Soils and Piling. Trial Bores. Design of Foundations. Design. The Proportions of Foundations for Engines, Turbines and Dynamo. Electric Machinery. Materials for Foundations. Holding Down Bolts and Anchor Plates. Excavation. Construction of Foundations. Vibration: Its Causes and Effects. Vibration. Methods of Isolating Machinery. The Fixing of Electric Motors.

Graphical Determination of Earth Slopes, Retaining Walls and Dams. **Charles Prelini, C.E.,** Author of “Tunnelling.” Demy 8vo. 8/6 net.

Rainfall, Reservoirs and Water Supply. **Sir Alexander Binnie, M.Inst.C.E., etc.** ([See p. 30.](#))

Timber. **J. E. Baterden, Assoc.M.Inst.C.E.** Extra Crown 8vo. 7/6 net.

Pricing of Quantities. Showing a Practical System of Preparing an Estimate from Bills of Quantities. **George Stephenson**, Author of “Repairs,” “Quantities,” etc. Extra Crown 8vo. 5/- net.

Building in London. A Treatise on the Law and Practice affecting the Erection and Maintenance of Buildings in the Metropolis. **Horace Cubitt, A.R.I.B.A.**, etc. Illustrated, with Diagrams. Royal 8vo. 31/6 net.

Theory and Practice of Designing. A progressive course of instruction in various branches of structural design. With numerous Illustrations. **Henry Adams, M.Inst.C.E., M.I.M.E.** 7/6 net.

“... We recommend it heartily, alike as a text-book or a handy work of reference.”—*Mechanical Engineer.*

Surveying, etc.

Adjustment of Observations by the Method of Least

Squares. With Application to Geodetic Work. **Professor Thomas Wallace Wright, M.A., C.E.,** and **John Fillmore Hayford, C.E.,** Chief of the Computing Staff U.S. Coast and Geodetic Survey. With Tables. 2nd Edn. Demy 8vo. 30/- net.

Surveying and Field Work. **J. Williamson, A.M.Inst.C.E.** Demy 8vo. 9/- net.

CONTENTS.—Surveying. Fundamental Principles. Chain Surveying Instruments—Chain Surveying. Field Operations—Chain Surveying. Running a Survey Line—Chain Surveying. Arrangement of Survey Lines—Chain Surveying. Errors—Chain Surveying. Special Problems. Plotting the Plan. Compass and Sextant Surveying. The Theodolite. Traverse Surveying with the Theodolite. Plotting a Traverse Survey by Angle and Distance. Plotting Traverse Survey by Co-ordinate or Latitude and Departure Method. Triangulation. Some Survey, Traverse and Triangulation Problems. Levelling. Errors in Levelling. Sections, Contours, etc. Setting out Curves, etc. Calculation of Areas. Calculation of Earthwork Quantities—Adjustment of Instruments. Appendix. Geometric and Trigonometric Formulæ. Index.

Practical Surveying. For Surveyors' Assistants and Students.

Ernest McCullough, C.E., Member of the American Society of Civil Engineers. Extra Crown 8vo. 229 Illustrations. 12/- net.

The Transition Spiral. **A. L. Higgins, A.R.C.S.** Crown 8vo. Illustrated. About 7/6 net.

Municipal Engineering.

The Efficiency of Pumps and Ejectors. **E. C. Bowden-Smith, M.I.Mech.E.**, etc. Late Egyptian Civil Service. Demy 8vo. Illustrated. ([See p. 17.](#)) 14/6 net.

Water Purification and Sewage Disposal. **Dr. J. Tillmans**, Director of the Chemical Department of the Municipal Institute of Hygiene, Frankfort-on-Maine. Translated by **Hugh S. Taylor, M.Sc.** 144 + xv pages. Demy 8vo. 9/- net.

Sewage Disposal Work: Design, Construction and Maintenance. Being a Practical Guide to Modern Methods of Sewage Purification. **Hugh P. Raikes, M.Inst.C.E.** Demy 8vo. 16/- net.

The Encyclopædia of Municipal and Sanitary Engineering. Edited by **W. H. Maxwell, A.M.Inst.C.E.**, Borough and Waterworks Engineer, Tunbridge Wells Corporation, etc. In one volume. Cloth. 42/- net.

Rainfall, Reservoirs and Water Supply. **Sir Alexander Binnie.** 9/- net.

CONTENTS.—Rainfall. Average. Fluctuations. Probable Average. Flow from the Ground. Intensity of Floods. Evaporation. Quantity and Rate per Head. Quality, Hardness, etc. Impurities. Filtration. Sources of Supply. Gravitation *versus* Pumping. Rivers and Pumping Works. Drainage Areas. Deductions from Rainfall. Compensation. Capacity of Reservoirs. Sites of Reservoirs. Puddle Trenches. Concrete Trenches. Base of Embankment. Reservoir Embankments. Puddle Wall. Formation of Embankment. Masonry Dams. Reservoir Outlets. Pipes through Embankment. Culvert under Embankment. Flow through Culverts. Valve Pit. Central Stopping. Tunnel Outlets. Syphon Outlets. Flood or Bye-channel. Waste Watercourse and Waste Weir. Aqueducts. Conduits. Pipes. Service Reservoir. Distribution. Valves. Meters. House Fittings. Index.

Conveyance and Distribution of Water for Water Supply. **Edward Wegmann, C.E.** Medium 8vo. 367 Illustrations. 8

Plates. 26/- net. (See page 22.)

Refuse Disposal and Power Production. **W. F. Goodrich.**

Demy 8vo. 16/- net.

Small Dust Destructors for Institutional and Trade

Refuse. **W. F. Goodrich.** Demy 8vo. 4/- net.

British Progress in Municipal Engineering. **William H.**

Maxwell, A.M.Inst.C.E. Super-Royal 8vo. 6/- net.

Water Pipe and Sewage Discharge Diagrams. **T. C.**

Ekin, M.Inst.C.E., M.I.Mech.E. Folio. 6/- net.

These diagrams and tables with accompanying descriptive letterpress and examples are based on Kutter's formula with a coefficient of roughness of 0.013 and give the discharges in cubic feet per minute of every inch diameter of pipe from 3 to 48 inches when running full on inclinations from 1 to 15 per 1000.

Water: its Purification and Use in the Industries. **W. W.**

Christie, M.Am.S.M.E., etc. 230 pages. 79 Illustrations. 12/6.

Modern Sanitary Engineering. **Gilbert Thomson, M.A.,**

M.Inst.C.E. Part I. HOUSE DRAINAGE. Demy 8vo. Vol. I. 2nd Edition. (Glasgow Text-Books of Civil Engineering.) 16/- net.

CONTENTS.—Introductory. The Site and Surroundings of the House. The General Principles of Drainage Design. Materials for Drains. The Size of Drains. The Gradient of Drains. Drain Flushing Traps: Their Principle and Efficacy. Traps: Their Number and Position. "Disconnections." Intercepting Traps and Chambers. Inspection Openings and Manholes. Soil, Waste and Connecting Pipes. Water Closets. Flushing Cisterns and Pipes. Urinals. Baths. Lavatory Basins. Sinks, Tubs, etc. Trap Ventilation. Designing a System of Drainage. Buildings of Special class. Test and Testing. Sanitary Inspections. Sewage Disposal for Isolated Houses. Index.

Part II. SEWERAGE. Demy 8vo. 18/- net.

CONTENTS.—General Principles of Design. Evolution of a Scheme. Size of Sewers. Quantity of Sewage. Velocity of Flow. Flushing. Excavations. Construction of Pipe Sewers. Construction of Built Sewers. Sea Outfall Sewers. Sewer Ventilation. Manholes. Lamp-holes. Street Gullies. Inverted Syphons. Raising Sewage. Storm Overflows. Connection to House Drains. Specifications and Schedules for Sewerage Work. Drainage and Sewerage of Housing Schemes. Index.

Town Planning in Madras. **H. V. Lanchester, F.R.I.B.A.,** etc.

12/6 net.

CONTENTS.—The Ethics of Town Planning. Historical Outlines of Town Planning. The Civic Survey. Economic Studies. Tradition and City Development. City Life and Housing. Commerce and Traffic. The Technique of City Development. Indian and European Cities. Problems in India. The Uses of the Town Planning Act. Madras City. The Improvement of Madras City.

Irrigation and Water Supply.

Water Purification and Sewage Disposal. **Dr. J. Tillmans.**
([See p. 30.](#))

Rainfall, Reservoirs and Water Supply, **Sir Alexander Binnie, M.Inst.C.E.** ([See p. 30.](#))

Irrigation: Its Principles and Practice as a Branch of Engineering. **Sir Hanbury Brown, K.C.M.G., M.Inst.C.E.** 2nd Edn. Revised. Demy 8vo. 16/- net.

CONTENTS.—Irrigation and its Effects. Basin Irrigation. Perennial Irrigation and Water “Duty.” Sources of Supply. Dams and Reservoirs. Means of Drawing on the Supply. Methods of Construction. Means of Distribution. Masonry Works on Irrigation Canals. Methods of Distribution of Water. Assessment of Rates and Administration. Flood Banks and River Training. Agricultural Operations and Reclamation Works. Navigation. Appendices. Index.

Irrigation Works. The “Vernon Harcourt” Lectures of 1910. **Sir Hanbury Brown, K.C.M.G., M.Inst.C.E.** Demy 8vo. Stiff Paper Cover. 1/- net.

Agriculture and Irrigation. **Kinsley D. Doyle.** Demy 8vo. 288 pp. 19/- net.

CONTENTS.—The Soil. Fertilizers. Irrigation. Wells and Pumping. Dry Farming and Economics. Pastures, Natural and Artificial. Notes on Forests, Cattle, Pigs and Carobs. Runs and Farms for Sheep and Ostriches. Cereals. Apples. Asparagus, Onions and Potatoes. Citrus Orchards. Grapes, Walnuts and Pistachio Nuts. Some Special Industries. Some Tropical Fruit. Tropical Plantations. Planting and Sowing for Oil. Tropical Agriculture.

The Practical Design of Irrigation Works. **W. G. Bligh, M.Inst.C.E.** New and Enlarged Edition in Preparation.

Notes on Irrigation Works. **N. F. Mackenzie, Hon. M.A., Oxon.; M.Inst.C.E.;** lately Under-Secretary for Irrigation to the Government of India. Demy 8vo. 7/6 net.

Telegraphy and Telephony.

Telegraph Engineering. For practising Telegraph Engineers and Engineering Students. It presents in a logical manner the subject of modern Land and Submarine Telegraphy from an Engineering viewpoint. **E. Hausmann, E.E., Sc.D.** Demy 8vo. 192 Illustrations. 12/6 net.

The Propagation of Electric Currents in Telephone and Telegraph Conductors. **J. A. Fleming, M.A., D.Sc., F.R.S.,** Pender Professor of Electrical Engineering in the University of London. 3rd Edn. Revised. Demy 8vo. 21/- net.

CONTENTS.—Preface. Mathematical Introduction. The Propagation of Electro-Magnetic Waves along Wires. The Propagation of Simple Periodic Electric Currents in Telephone Cables. Telephony and Telegraphic Cables. The Propagation of Currents in Submarine Cables. The Transmission of High-Frequency and Very Low-Frequency Currents along Wires. Electrical Measurements and Determination of the Constants of Cables. Cable Calculations and comparison of Theory with Experiment. Loaded Cables in Practice.

Toll Telephone Practice. **J. B. Theiss, B.S., LL.B.,** and **Guy A. Joy, B.E.** With an Introductory Chapter by Frank F. Fowle, S.B. 14/- net.

CONTENTS.—Rural Telephone Equipment. Toll Cut-in Stations. Toll Positions at a Local Switchboard. Toll Switching Systems. Small Toll Switchboards. Multiple-Drop Toll Switchboards. Multiple-Lamp Toll Switchboards. Toll Connections to Local Automatic Systems. Supervisory Equipment and Toll Chief Operator's Desk. Toll Wire Chief's Desk. Simplex Systems. Composite Systems. Phantom Lines. Test and Morse Boards. Small Test Panels. Line Construction. Electrical Reactions in Telephone Lines. Cross and Talk Inductive Disturbances. Methods of Testing. Toll Line Maintenance. The Telephone Repeater.

Radio-Telegraphy. **C. C. F. Monckton, M.I.E.E.** With 173 Diagrams and Illustrations. Extra Crown 8vo. 7/6 net.

Radiographic Technique. **T. Thorne Baker, F.C.S., F.R.P.S., A.I.E.E.** Demy 8vo. 208 pp. Illustrated. 15/-.

CONTENTS.—The Scope of X-Ray Work. The Photographic Plate. The Storage and Handling of Plates. Intensifier Screens. Fluoroscopic Examination. Localisation by Radiography. The Dark Room and its Equipment. Intensification and Reduction. The Technique of the Photographic Print. Industrial Applications of the X-Rays. Analysis with the X-Rays. Appendix.

The Telegraphic Transmission of Photographs. **T. Thorne Baker, F.C.S., F.R.P.S., A.I.E.E.** With 66 Illustrations and Diagrams. Crown 8vo. 2/6 net.

Telegraphy, Aeronautics and War. **Charles Bright, F.R.S.E., M.Inst.C.E.,** etc., Author of “Submarine Telegraphs,” “Imperial Telegraphic Communication,” etc. Demy 8vo. 16/- net.

Electrical Engineering.

Single-Phase Electric Railways. **E. Austin.** Over 290 pages.
346 Illustrations. 21/- net.

CONTENTS.—The Single-Phase System. The London, Brighton and South Coast Railway. The Midland Railway. The Midi Railway. The Blankenese-Hamburg-Ohlsdorf Railway. The Dessau-Bitterfeld Railway. The Murnau-Oberammergau Railway. The Mittenwald Railway. The St. Polton-Mariazell Railway. The Martigny-Orsières Railway. The Valle-Maggia Railway. The Rhætian Railway. The Lötschberg-Simplon Railway. The Rotterdam-Scheveningen Railway. The Thamshavn-Lokken Railway. The Rjukan Railway. The Swedish State Railways. The Parma Single-Phase Tramways. The New York, New Haven and Hartford Railway. The New York, Westchester and Boston Railway. The Spokane and Inland Empire Railway. The Rock Island and Southern Railway. The Hoosac Tunnel Railway. The St. Clair Tunnel Railway.

Application of Electric Power to Mines and Heavy Industries. **W. H. Patchell, M.Inst.C.E., M.I.M.E., M.Am.I.E.E.**
([See p. 23.](#))

Overhead Transmission Lines and Distributing Circuits: Their Design and Construction. A very thorough and much-needed Treatise dealing exhaustively with this important subject. **F. Kapper.** Translated by **P. R. Friedlaender, M.I.E.E.** Crown 4to. Fully Illustrated. 16/- net.

CONTENTS.—Conductor Materials. Sag and Tension of the Line. Design of the Supporting Structures. Stability of Poles and Masts. The Forces on the Foundation Piece. Fixing the Poles in the Ground. Concrete and Cement Foundation Work. Pole Construction. Overhead Line insulators. Attachment of the Insulators to their Pins. Attachment of the Wire to the Insulators. Joints and Branch Connections on the Wire. Arrangement of the Wires. Earthing. Crossing over Postal Wires, Railways and Roads. Erection of Poles and Masts. Erecting the Wire. Rules and hints for the Design and Erection of Overhead Lines. Instruments for Surveying and Laying out the Route of a Line. Surveying the Route of a Transmission Line. The Most Economical Length of Span. Comparison of the various Supporting Structures with Reference to the Minimum Annual Charges. Local Overhead Distributing Systems. Agreements with Contractors. Tools and Appliances. Regulations dealing with the Erection and Operation of Overhead Lines.

Electric Mine Signalling Installations. A Practical Treatise on the Fitting-up and Maintenance of Electrical Signalling Apparatus

in Mines. **G. W. Lummis Paterson.** Crown 8vo. 140 Illustrations. 5/- net.

CONTENTS.—Ringing Keys and Tappers. Electric Mining Bells, Signal Alarms and Relays. Electric Mining Indicators or Annunciators. Electric Generators for Mine Signal Installations. Primary Batteries for Signal Installations. Electric Mining Shaft Signals. Electric Shaft Signal Wiring. Electric Engine Plane Signals. Electric Engine Plane Signal Wiring. Appendix.

Electric Welding and Welding Appliances. **H. Carpmael.** Imperial 8vo. Illustrated. (The Engineer Series.) 18/- net.

CONTENTS.—Introduction. The Bernardos Carbon Arc Process. Resistance Welding. Arc Welding at Steel Barrel Works. The Pontelec Methods and Machines. The Quasi-Arc Process. Resistance Welders of the British Insulated and Helsby Cables, Ltd. Machines and Apparatus for Arc Welding. Resistance Welders of the Electric Welding Co., Ltd. Some Large American “Spot” Welders. Oil Drum Making by Resistance Welding. Resistance Welders of the A1 Manufacturing Co., Ltd. The strength of Electric Welds. A large British “Spot” Welding Machine. The “Plastic-Arc” Welding System. The “A.C.” System of Arc Welding. Index.

Transmission Line Formulas. For Electrical Engineers and Engineering Students. **H. B. Dwight, A.M.Am.I.E.E.** 143 pages. 28 Diagrams. 12/6 net.

Electric Cooking, Heating, Cleaning, etc. Being a Manual of Electricity in the Service of the Home. “**Housewife**” (**Maud Lancaster**). Edited by **E. W. Lancaster, A.M.Inst.C.E., M.Inst.E.E.** Fully Illustrated. 338 + x pages. Demy 8vo. 4/- net.

Electric Mechanism. Part I. SINGLE-PHASE COMMUTATOR MOTORS. **F. Creedy, A.C.G.I., A.M.I.E.E., Assoc. Am.I.E.E.** Demy 8vo. 7/6 net.

Industrial Electrical Measuring Instruments. **Kenelm Edgcumbe, M.Inst.C.E., M.I.E.E.** Demy 8vo. 430 pages. 260 Illustrations. 16/- net. 2nd Edition, entirely Re-written and Enlarged.

CONTENTS.—Errors and Accuracy. Constructional Details. Measurement of Resistances and Insulation. Potentiometers. Standard Methods of Current and Pressure Measurement. Moving Iron. Moving Coil. Hot Wire. Induction and Dynamometer Ammeters and Voltmeters. Electrostatic Voltmeters. Power Measurement. Dynamometer, Induction, Electrostatic and Hot-wire Wattmeters. Idle or Reactive Component Meters. Phase or Power Factor Meters. Phase Rotation Indicators. Frequency Meters. Fault and Leakage Detectors. Synchronising Devices. Pyrometers. Instrument Transformers. Graphic or Recording Instruments. Electrical Speed Indicators. Oscillograph and other Wave Form Indicators. Live Main or Charge Indicators. Earth-plate Testing Sets. Rail-bond Testing Sets. Cell-testing Voltmeters. Motor-Car Voltmeters and Ammeters. Voltage and Current Relays.

Measurement of the Starting Current of a Motor. Paralleling. Compensated and Crest or Peak Voltmeters.

Calculation of Electrical Conductors. **W. T. Taylor, M.Inst.C.E.** Imperial 8vo. 32 pages. Diagrams in the Text. Chart at the end. 10/6 net. The Chart sold separately, 2/- net.

Solenoids, Electromagnets and Electromagnetic Windings. **Charles Underhill, A.M.A.I.E.E.** 346 pages. 218 Illustrations. Demy 8vo. 2nd Edn. Revised. 10/6 net.

Switches and Switchgear. **R. Edler.** Translated, Edited and Adapted to English and American Practice by **Ph. Laubach, A.M.I.E.E.** Fully Illustrated. Demy 8vo. 16/- net.

CONTENTS.—General Remarks on the Design of Switchgear. Connecting Leads. Cable Sockets. Connections. Copper Bars. Contact Blocks. Bolts. Contact Springs and Brushes. Carbon Contacts. Devices to Eliminate Sparking at the Main Contacts. Switches and Change-over Switches for Low Pressure and Medium Pressure. High Pressure Switches. Fuses. Self-Acting Switches (Automatic Switches). Starting and Regulating Resistances. Controllers.

High and Low Tension Switchgear Design. **A. G. Collis, A.M.I.E.E.** Demy 8vo. 94 Illustrations. 12/- net.

Foster's Electrical Engineer's Pocket Book. A Handbook of Useful Data for Electricians and Electrical Engineers. Edited by Horatio **A. Foster, M.A.I.E.E., M.A.S.M.E.** With the Collaboration of Eminent Specialists. 5th Edition, completely Revised and Enlarged. Pocket size. Leather bound. With thumb index. Over 1600 pages. Illustrated. 36/- net.

The Propagation of Electric Currents in Telephone and Telegraph Conductors. **J. A. Fleming, M.A., D.Sc., F.R.S.** 21/- net. ([See p. 32.](#))

Dynamo Laboratory Manual. **W. S. Franklin and William Esty.**

Vol. I. DIRECT-CURRENT STUDIES AND TESTS. 12/- net.

Elementary Electricity and Magnetism. **W. S. Franklin** and **Barry MacNutt.** 9/- net.

The Electrical Conductivity and Ionization Constants of Organic Compounds. **Heyward Scudder, B.A., B.S., M.D.** Royal 8vo. 568 pages. 17/6 net.

Electricity: A Text-book of Applied Electricity. **H. M. Hobart, B.Sc., M.Inst.C.E.** With 43 Tables and 115 Illustrations. Demy 8vo. 6/6 net.

Advanced Theory of Electricity and Magnetism. **W. S. Franklin** and **Barry MacNutt.** 16/6 net.

Elements of Electricity and Magnetism. **W. S. Franklin** and **Barry MacNutt.** 12/- net.

The Elements of Electrical Engineering. **W. S. Franklin.**

Vol. I. DIRECT AND ALTERNATING CURRENT MACHINES AND SYSTEMS. 28/- net.

Vol. II. ELECTRIC LIGHTING AND MISCELLANEOUS APPLICATIONS OF ELECTRICITY. 20/- net.

Heavy Electrical Engineering. **H. M. Hobart, B.Sc., M.Inst.C.E.** Illustrated with Diagrams, etc. Demy 8vo. 16/- net.

Electric Railway Engineering. **H. F. Parshall, M.Inst. C.E.,** etc., and **H. M. Hobart, B.Sc., M.Inst.C.E.** 476 pages and nearly 600 Diagrams and Tables. Imperial 8vo. 42/- net.

American Electric Central Station Distribution Systems. **Harry Barnes Gear, A.M.A.I.E.E.,** and **Paul Francis Williams, A.M.A.I.E.E.** Demy 8vo. 2nd Edn. Revised and Enlarged. 18/- net.

Hydroelectric Developments and Engineering. A Practical and Theoretical Treatise on the Development, Design, Construction, Equipment and Operation of Hydroelectric Transmission

Plants. **Frank Köester.** 479 pages. 500 Illustrations. 45 Tables. Imperial 8vo, 42/- net.

Steam-Electric Power Plants. A Practical Treatise on the Design of Central Light and Power Stations and their Economical Construction and Operation. **Frank Köester.** 474 pages. Fully Illustrated. Crown 4to. 31/6 net.

Electric Power and Traction. **F. H. Davies, A.M.I.E.E.** With 66 Illustrations. Extra Crown 8vo. 7/6 net.

Laboratory and Factory Tests in Electrical Engineering. **George F. Sever and Fitzhugh Townsend.** 2nd Edn. Thoroughly revised. 282 pages. Demy 8vo. 10/6 net.

The Theory of Electric Cables and Networks. **Alexander Russell, M.A., D.Sc.** 2nd Edn. Entirely Revised and largely Rewritten. 24/- net.

Testing Electrical Machinery. **J. H. Morecroft and F. W. Hehre.** 15/- net.

Continuous Current Engineering. **Alfred Hay, D.Sc., M.I.E.E.** 2nd Edn. A New and Revised Edition of this Popular Text-Book. 30 pages. 183 Illustrations. Demy 8vo. 9/6 net.

CONTENTS.—Electrical Units. Electromagnets. Magnetism. Hysteresis. Measuring Instruments. Dynamo Construction. Dynamo Used as Motor. Construction and Management of Motors. Secondary Cells and Their Uses. The Electric Arc. Mercury Vapour Lamps. Incandescent Lamps. Photometry. Switches. Conductors. Insulation.

Direct and Alternating Current Testing. **Frederick Bedell, Ph.D.,** Professor of Applied Electricity, Cornell University. Assisted by **Clarence A. Pierce, Ph.D.** Enlarged and Revised Edn. Demy 8vo. 11/- net.

Internal Wiring of Buildings. **H. M. Leaf, M.I.M.E.,** etc. 3rd Edn. Crown 8vo. 3/6 net.

Electric Lamps. **Maurice Solomon, A.M.I.E.E.** 7/6 net.

CONTENTS.—The Principles of Artificial Illumination. The Production of Artificial Light. Photometry. Methods of Testing. Carbon-Filament Lamps. The Nernst Lamp. Metallic-Filament Lamps. The Electric Arc. The Manufacture and Testing of Arc Lamp Carbons. Arc Lamps. Miscellaneous Lamps. Comparison of Lamps of Different Types.

The Range of Electric Searchlight Projectors. **Jean Rey,** Ingenieur Civil des Mines Lauréat de l'Institut. Authorised translation by **J. H. Johnson, M.Inst.E.E., Assoc. A.I.E.E.** 34 Illustrations and Diagrams. 12/6 net.

CONTENTS.—Part I: Illumination by Electric Searchlight Projectors. Determination of the Luminous Flux Emitted by an Electric Arc. Theoretical Illumination obtained with a Specified Reflector. Efficiency of Electric Searchlight Projectors. Practical Value of the Illumination from Searchlight Projectors. Searchlights with Mangin Mirrors. Part II: Range of Electric Searchlight Projectors. General Remarks. General Law giving the Value of the Range apart from the Visual Sharpness. Practical Application of the Law of Ranges, apart from the Visual Acuity. Practical Examples and Problems in respect of the Range. The Variation in the Range with the Size of the Searchlight and the Atmospheric Transparency. Visual Acuity of Sharpness. Appendices and Index.

Searchlights: Their Theory, Construction and Application. **F. Nerz.** Translated from the German by **Charles Rodgers.** Royal 8vo. 7/6 net.

Essentials of Electrical Engineering. **John Fay Wilson.** Royal 8vo. 282 Illustrations. 16/- net.

CONTENTS.—The Electric Circuit. Magnetism and Magnetic Induction. Practical Construction of the Dynamo. The Continuous Current Generator. The Continuous Current Motor. Losses, Efficiencies and Ratings of Continuous Current Dynamos. Polyphase Alternating Currents. The Alternating Current Generator. The Synchronous Motor. Current Rectifying Apparatus. The Transformer. Transformer Connections. The Induction Motor. Single-phase Commutating Motors. Electric Lamps. Circuit Interrupting Apparatus. Meters. Power Transmission and Distribution. The Storage Battery. Harmonic Quantities. Inductance. Capacity. The Complex Quantity. Admittance, Conductance and Susceptance. Resuscitation from Electric Shock.

Electric Wiremen's Manuals. ([See p. 7.](#))

Electro-Chemistry, etc.

Electro-Metallurgy. **J. B. C. Kershaw, F.I.C.** With 61 Illustrations.
Extra Crown 8vo. 7/6 net.

CONTENTS.—Aluminium. Bullion and Gold. Calcium Carbide and Acetylene Gas. Carborundum. Copper. Ferro-Alloys. Glass and Quartz Glass. Graphite. Iron and Steel. Lead. Miscellaneous Products. Nickel. Sodium. Tin. Zinc.

The Manufacture of Chemicals by Electrolysis. **A. J. Hale, B.Sc., F.I.C.** (A Treatise of Electro-Chemistry. Edited by **Bertram Blount, F.I.C.**) Demy 8vo. Illustrated. 6/- net.

CONTENTS.—Electrolytic Hydrogen and Oxygen. Ozone. Production of Per-Salts and Hydrogen Peroxide. Nitric Acid. Hydroxylamine. Hydrosulphites. Fluorine. Electrolytic Preparation of Pigments and Insoluble Substances. Electro-Osmotic and Electro-Colloidal Processes. Electrolytic Reduction of Organic Compounds. Oxidation and Substitution of Organic Compounds. Appendix. Name Index. Subject Index.

Ozone. **S. Rideal, D.Sc., F.I.C., F.C.S.** (A Treatise of Electro-Chemistry. Edited by **Bertram Blount, F.I.C.**) Demy 8vo. 12/- net.

CONTENTS.—Early History of Ozone and its General Properties. The Natural Occurrence of Ozone. Chemical Production. Thermal Production. The Electrolytic Preparation of Ozone. Production by Ultra-Violet Radiation and by Ionic Collision. Production by means of the Silent Electric Discharge. The Catalytic Decomposition of Ozone. Industrial Applications. Methods of Detection and Analysis. Name Index. Subject Index.

Lighting.

Electric Lamps. Maurice Solomon, A.C.G.I., A.M.I.E.E. Extra Crown 8vo. 7/6 net. ([See p. 38.](#))

Town Gas and its Uses for the Production of Light, Heat, and Motive Power. W. H. Y. Webber, C.E. With 71 Illustrations. Extra Crown 8vo. 7/6 net.

Thermodynamics.

Entropy: Or, Thermodynamics from an Engineer's Standpoint, and the Reversibility of Thermodynamics. **James Swinburne, M.Inst.C.E., M.I.E.E.,** etc. Illustrated with Diagrams. Crown 8vo. 5/- net.

Applied Thermodynamics for Engineers. **William D. Ennis, M.E., M.Am.Soc.M.E.,** Professor of Mechanical Engineering in the Polytechnic Institute of Brooklyn. With 316 Illustrations. Royal 8vo. 21/- net.

Technical Thermodynamics. **Dr. Gustav Zeuner.** First English Edition. From the Fifth Complete and Revised Edition of "Grundzuge der Mechanischen Wärmetheorie."

Vol. I. FUNDAMENTAL LAWS OF THERMODYNAMICS; THEORY OF GASES.

Vol. II. THE THEORY OF VAPOURS.

Authorised translation by **J. F. Klein, D.E.,** Professor of Mechanical Engineering, Lehigh University. Fully Illustrated. 2 Volumes. Demy 8vo. 42/- net.

A Text-Book of Thermodynamics. With special reference to Chemistry. By **J. R. Partington, M.Sc.,** of the University of Manchester. 544 + x pages. 91 Diagrams. Demy 8vo. 10/6 net.

CONTENTS.—Thermometry and Calorimetry. The first law of Thermodynamics and some Applications. The Second Law of Thermodynamics; Entropy. The Thermodynamic Functions and Equilibrium. Fluids. Ideal and Permanent Gases. Changes of Physical State. Van Der Waals' Equation and the Theory of Continuity of States. Thermochemistry. Gas Mixtures. The Elementary Theory of Dilute Solutions. Chemical Equilibrium in Gaseous Systems. Equilibrium in Dilute Solutions. General Theory of Mixtures and Solutions. Capillarity and Adsorption. Electrochemistry. The Theorem of Nernst. Kinetic Theories in Thermodynamics. Index.

Experimental and Theoretical Applications of Thermodynamics to Chemistry. **Professor Walter Nernst,** University of Berlin. Extra Crown 8vo. 9/- net.

Principles of Thermodynamics. **G. A. Goodenough, M.E.** 2nd
Edn. Revised. 14/- net.

Physics and Chemistry.

The Measurement of Steady and Fluctuating Temperatures. **R. Royds, M.Sc., A.M.I.Mech.E.** Demy 8vo. 162 pp.

This book gives a concise account of modern methods of measuring both steady and fluctuating temperatures and of their applications in industrial practice.

Heat Transmission. **R. Royds, M.Sc., A.M.I.Mech.E.** Demy 8vo.

Vol. I. BY RADIATION, CONDUCTION, AND CONVECTION.

Vol. II. IN BOILERS, CONDENSERS, AND EVAPORATORS.

Each volume sold separately.

Standards and Tests for Reagent Chemicals. **Benjamin L. Murray.** Member of the American Chemical Society, Chemical Society (London), Society of Chemical Industry, etc. etc. 385 pages. Demy 8vo.

The Silicates in Chemistry and Commerce. **Drs. W. and D. Asch.** Translated with Critical Notes and some Additions by **A. B. Searle.** 400 + xx pages. Royal 8vo. 21/- net.

CONTENTS.—The Chemistry of Carbon and Silicon. Historical Review of Existing Theories concerning the Constitution of the Alumino-silicates and other Silicates. Critical Examination of Existing Theories concerning Alumino-silicates. A Hypothesis concerning the Bonding of the Atoms in Alumino-silicates and Allied Compounds. The Consequences of the “Hexite-Pentite Theory,” and the Facts: (I.) The Reaction during Double Decomposition; (II.) The Genetic Relationship between the various Alumino-silicates; (III.) The Possibility of a Chemical System of Alumino-silicates; (IV.) The Variable Chemical Behaviour of part of the Aluminium in Kaolin, Nepheline and in the Epidotes; (V.) The Minimum Molecular Weight of Alumino-silicates; (VI.) The Constitution of Andesite; (VII.) The Possibility of Isomerism; (VIII.) Water of Crystallisation and of Constitution: Basic and Acid Water; (IX.) Prognoses; (X.) The Constitution of the Complexes of Molybdenum and Tungsten; (XI.) The Constitution of Clays; (XII.) Ultramarines; (XIII.) A New Theory of Hydraulic Binding Materials and particularly of Portland Cements; (XIV.) A New Theory of the Porcelain Cements as used for Dental Fillings; (XV.) A New Theory of Glass, Glazes and Porcelain; (XVI.) The Hexite-Pentite Theory as a General Theory of Chemical Compounds. The Conversion of the H.P. Theory into a Stereochemical Theory and the combination of the latter with the modern theory of the

Structure of Crystals. Summary and Conclusions. Bibliography of references mentioned in Text. Appendix. Formulæ and Analyses. Bibliography of references in Appendix.

Industrial Chemistry. A Manual for the Student and Manufacturer. 2nd Edn. Revised and Enlarged. Edited by **Allen Rogers**, in charge of Industrial Chemistry, Pratt Institute, Brooklyn, N.Y.; in collaboration with many Experts. 340 Illustrations. Royal 8vo.

CONTENTS.—General Processes. Materials of Construction. Water for Industrial Purposes. Fluids. Producer Gas. Power Transmission, Boilers, Engines and Motors. Sulphuric Acid. Nitric Acid. Salt and Hydrochloric Acid. Commercial Chemicals. Chlorine and Allied Products. Electro-chemical Industries. Lime, Cement and Plaster. Clay, Bricks and Pottery. Glass. Dutch Process White Lead. Sublimed White Lead. Pigments, Oils and Paints. The Metallurgy of Iron and Steel. Fertilizers. Illuminating Gas. Coal Tar and its Distillation Products. The Petroleum Industry. The Destructive Distillation of Wood. Oils, Fats and Waxes. Lubricating Oils. Soaps, Glycerine and Candles. Laundering. Essential Oils, Synthetic Perfumes and Flavouring Materials. Resins, Oleo-Resins, Gum-Resins and Gums. Varnish. Sugar. Starch, Glucose, Dextrin and Gluten. Brewing and Malting. Wine Making. Distilled Liquors. Textiles. Dyestuffs and their Application. The Art of Paper Making. Explosives. Leather. Vegetable Tanning Materials. Glue and Gelatine. Casein.

Elements of Industrial Chemistry. Edited by **Allen Rogers**. An Abridgment of the Manual of Industrial Chemistry prepared especially for Students. Demy 8vo. 520 pages. 117 Illustrations. 16/- net.

CONTENTS.—General Processes. Water for Industrial Purposes. Fuels. Sulphuric Acid. Nitric Acid. Hydrochloric Acid. Elements and Compounds. Electro-chemical Industries. Lime, Cement and Plaster. Ceramic Industries. Pigments and Paints. Iron and Steel. Fertilisers. Organic Chemicals. Illuminating Gas. Coal Tar and Distillation Products. Petroleum. Distillation of Wood. Oils, Fats and Waxes. Soap and Glycerine. Essential Oils. Gums and Resins. Sugar, Starch and Glucose. Brewing. Wine Making and Distilling. Textiles. Dyestuffs. Paper and Cellulose Industries. Explosives. Leather.

Laboratory Guide of Industrial Chemistry. **Allen Rogers**, Editor of “A Manual of Industrial Chemistry,” “Elements of Industrial Chemistry,” etc. Demy 8vo. 212 pages. 33 Illustrations. 2nd Revised and Enlarged Edition. 15/- net.

CONTENTS.—General Process. Inorganic Preparations. Organic Preparations. Dyeing of Textile Fibres. Pigments and Lakes. Driers, Varnishes, Paints and Stains. Soap and Allied Products. Leather Manufacture. Wood Fibre, Pulp and Paper. Useful Data.

The Application of Physico-Chemical Theory to Technical Processes and Manufacturing Methods. **Dr. R. Kremann.** Translated by **H. E. Potts, M.Sc.** ([See p. 51.](#))

The Chemistry and Technology of Printing Inks.

Norman Underwood and **Thomas V. Sullivan**. Medium 8vo. 12/6 net.

Experimental and Theoretical Applications of Thermodynamics to Chemistry. **Walter Nernst**,

University of Berlin. Extra Crown 8vo. 9/- net.

Vapours for Heat Engines. **William D. Ennis, M.E.,**

M.Am.Soc.M.E., Professor of Mechanical Engineering in the Polytechnic Institute of Brooklyn, Author of "Applied Thermodynamics for Engineers," etc. With 21 Tables and 17 Illustrations. Demy 8vo. 6/6 net.

The Manufacture of Chemicals by Electrolysis. A

Treatise of Electro-Chemistry Series. **A. J. Hale, B.Sc., F.I.C.** Demy 8vo. Illustrated. 6/- net. ([See p. 39.](#))

The Hydrogenation of Oils. Catalyzers and Catalysis and the

Generation of Hydrogen. **Carleton Ellis, S.B.** New and Revised Edition. 40/- net.

Introduction to the Chemistry and Physics of

Building Materials. **Alan E. Munby, M.A.(Cantab.).** Extra Crown 8vo. 7/6 net.

A Calendar of Leading Experiments. **W. S. Franklin** and

Barry Macnutt. 16/6 net.

The Chemistry of Paints and Paint Vehicles. By **C. H.**

Hall, B.S. Crown 8vo.

The Chemistry of the Oil Industries. **J. E. Southcombe,**

M.Sc., Lecturer on Oils and Fats, Royal Salford Technical Institute, etc. Illustrated. 9/- net.

CONTENTS.—Preface. Introductory Organic Chemistry. Mineral Oils. Natural Sources and Methods of Preparation of the Saponifiable Oils and Fats. Impurities occurring in Crude Oils and Fats and the Technical Methods of Removing them. Composition and Properties of the Saponifiable Oils and Fats in general. The Natural Waxes. Analytical Methods. Industrial Applications of Fats and Oils. Saponification of Fats and Oils. Candle Manufacture, Soap Making, Glycerine. Conclusion. Scientific and Technical Research on Problems in the Oil and Related Industries. Literature Index. Illustrations

The Chemistry of the Coal Tar Dyes. **Irving W. Fay, Ph.D.**
(Berlin). Demy 8vo. 470 pages. 21/- net.

The Chemistry of the Rubber Industry. **Harold E. Potts,**
M.Sc., Member International Rubber Testing Committee. Demy 8vo.
7/6 net. 2nd Impression.

CONTENTS.—The Colloidal State. Raw Rubber. Gutta-Percha and Balata. Mixing. Vulcanisation, History. Vulcanised Rubber.

The Chemistry of Cement, Concrete, and Bricks. **A. B.**
Searle. Demy 8vo. 10/6 net.

CONTENTS.—The Raw Materials for Cements. Methods of Cement Manufacture. The Chemical and Physical Changes in Cements. The Changes which occur in Setting and Hardening. Testing the Properties of Cements. The Components of Concrete and their Properties. The Preparation of Concrete. Reinforced Concrete. Special Properties of Concrete. Testing Concrete. The Raw Materials for Bricks. Methods of Brickmaking. The Chemical and other Changes in Drying and Burning Bricks. The Properties of Bricks. Siliceous Bricks. Basic and Neutral Bricks. Index.

Liquid Air and the Liquefaction of Gases. **T. O'Connor**
Sloane, M.A., M.E., Ph.D. Demy 8vo. 2nd Edn. 21/- net.

Detection of the Common Food Adulterants. A handbook
for health officers, food inspectors, chemistry teachers and students.
Prof. E. M. Bruce. Crown 8vo. 2nd Edn. Revised. 10/6 net.

The Chemistry of Dyeing and Bleaching of Vegetable
Fibrous Materials. **Julius Hübner, M.Sc.Tech., F.I.C.** Demy
8vo. 2nd Impression. Very fully Illustrated. 20/- net.

CONTENTS.—The Vegetable Fibres. Water. Chemicals and Mordants. Bleaching. Mercerising. Mineral Colours. The Natural Colouring Matters. Basic Cotton Dyestuffs. Substantive Cotton Dyestuffs. Sulphur Dyestuffs. Acid and Resorcine Dyestuffs. Insoluble Azo-Colours, produced on the fibre. The Vat Dyestuffs. Mordant Dyestuffs. Colours produced on the fibre by Oxidation. Dyeing Machinery. Estimation of the Value of Dyestuffs. Appendix. Index.

The Colloidal and Crystalloidal State of Matter. **H. E. Potts, M.Sc., and W. J. Breitland.** Translated from the German of Professor **P. Rohland.** Crown 8vo. 4/6 net.

Colloids in Biology and Medicine. **Prof. H. Beechold.** Translated from 2nd German Edition by **J. M. Bullaya, A.B., M.D.** Illustrated. 31/6 net.

The Nitrocellulose Industry. A compendium of the history, chemistry, manufacture, commercial application and analysis of nitrates, acetates and xanthates of cellulose as applied to the peaceful arts, with a chapter on gun cotton, smokeless powder and explosive cellulose nitrates. **Edward Chauncey Warden, Ph.C., M.A., F.C.S.** 2 Volumes. 1240 pages. 324 Illustrations. Small 4to. 50/- net.

Trinitrotoluenes and Mono- and Dinitrotoluenes: Their Manufacture and Properties. **G. Carlton Smith, B.Sc.** Extra Crown 8vo. 140 pages. 8/6 net.

CONTENTS.—Introduction. Historical. Theory of the Nitration of Toluene. The Manufacture of T.N.T. The Purification of T.N.T. Inspection and Testing. Properties of the Trinitrotoluenes, Mono- and Dinitrotoluenes. Accidents in T.N.T. Plants. Diseases.

Materials for Permanent Painting. A manual for Manufacturers, Art Dealers, Artists and Collectors, explaining the composition of the materials used in painting. **M. Toch, F.C.S.** Frontispiece in Colour and other Illustrations. Crown 8vo. 8/- net.

Principles of Quantitative Analysis. **Walter C. Blasdale, Ph.D.** Demy 8vo. 400 pages. 70 Illustrations. 17/- net.

CONTENTS.—Introductory Statements and Definitions. General Features of Gravimetric Processes. Gravimetric Gas Evolution Processes. Gravimetric Precipitation Processes. Gravimetric Solution and Extraction Processes. Partition Processes. General Features of Volumetric Processes. Volumetric Processes Involving Precipitation. Volumetric Processes Involving Neutralisation. Volumetric Processes Involving Oxidation. Physico-Chemical Processes.

The Theory and Use of Indicators. An Account of the Chemical Equilibria of Acids, Alkaloids and Indicators in Aqueous

Solution, with Applications. **E. B. R. Prideaux, M.A., D.Sc.** Demy 8vo. 390 pages. 12/6 net.

CONTENTS.—The Equilibria of Acids, Bases and Salts, and the Physical Methods of Determining Acidity and Alkalinity. Light Absorption in the Visible Spectrum and Colorimetry. Theories of Colour in their Relation to the Ionic Theory, Chemical Constitution and the Formation of Salts. The Colour of Indicators as a Function of Hydrion Concentration. The Determination and Use of Indicator Constants. The Preparation and Use of Solutions of Standard Hydrion Concentration. Applications. The Course of Neutralisation and the Theory of Titration. The Solution Equilibrium and Titration of some Acids. A Summarised List of the Principal Indicators with Absorption Spectra.

Problems in Physical Chemistry. With Practical Applications. **E. B. R. Prideaux, M.A., D.Sc.**, with a Preface by **F. G. Donnan, M.A., Ph.D., F.R.S.** 2nd Edn. Revised. 18/- net.

CONTENTS.—Preface. Author's Preface. Mathematical Methods and Formulæ. Table of Logarithms. List of Symbol and Abbreviations. Units and Standard of Measurement. Thermochemistry. Systems of One Component. Systems of Two Components. Reactions in Solution. Electromotive Force and Chemical Reaction. Velocity of Chemical and Radioactive Changes. Reactions in Gases.

Chemical Theory and Calculations. An Elementary Text-Book. **Forsyth J. Wilson, D.Sc.(Edin.), Ph.D.(Leipzig)**, and **Isidor M. Heilbron, Ph.D. (Leipzig), F.I.C., A.R.T.C.** 2nd Edition. 4/6 net.

Chemical Calculations. **R. Harman Ashley, Ph.D.** Extra Crown 8vo. 10/- net.

CONTENTS.—Ratios. Approximate Numbers. Interpolation. Heat. Specific Gravity. Gas Calculations. Calculation of Atomic Weights and Formulas. Gravimetric Analysis. Volumetric Analysis. Use of Specific Gravity Tables and Acid Calculations.

The Nature of Solution. **H. C. Jones**, with a Biographical Memoir by **Professor E. E. Reid.** Large 8vo. 400 pages. 18/- net.

CONTENTS.—Importance of Solution. Earlier Views as to the Nature of Solution. The Osmotic Pressure of Solutions. Relations between Solutions and Gases Demonstrated by Vant Hoff. The Theory of Electrolytic Dissociation as Announced by Arrhenius. Diffusion in Solution. Depression of the Vapour-tension of a Solvent by Substances Dissolved in it. Depression of Freezing-Point of a Solvent by the Solute. Aqueous Solutions of Acids, Bases and Salts—Electrolytes. Some Electrical Properties of Aqueous Solutions of Electrolytes. Solution in Non-aqueous and in Mixed Solvents. Colloidal Solutions. Solutions in Solids as Solvents. The Newer Hydrate Theory. The Solvate Theory of Solutions.

A Text-Book of Thermodynamics. With Special Reference to Chemistry. **J. R. Partington, M.Sc.** 10/6 net. ([See p. 40.](#))

Text-Book of Elementary Chemistry. **F. Mollwo Perkin, Ph.D., F.I.C., F.C.S.** and **Eleanor M. Jagers.** Crown 8vo. 4/- net.

CONTENTS.—Examination of Water. Metric System. Distillation. Thermometers. Freezing Points. Boiling Points. Melting Points. The Atmosphere. Chemical Properties of the Atmosphere. Chemical Properties of Water. Hydrogen and Oxygen. Quantitative Examination of Chalk. Carbon and its Oxides and Compounds with Hydrogen. Flame and Combustion. Compounds of Nitrogen. Hydrochloric Acid and the Halogens. Sulphur and its Compounds. Phosphorus and Phosphorous Compounds. The Metals. Technical Processes. Appendix. Index.

Practical Methods of Inorganic Chemistry. **F. Mollwo Perkin, Ph.D.** Crown 8vo. 3/6 net.

The Identification of Organic Compounds. **G. B. Neave, M.A., D.Sc.,** and **I. M. Heilbron, Ph.D., F.I.C.,** etc. Crown 8vo. 2nd Edn. 4/6 net.

Contemporary Chemistry. **E. E. Fournier D'Albe, B.Sc., A.R.C.S., M.R.I.A.,** Author of "The Electron Theory," etc. 4/- net.

A New Era in Chemistry. Some of the more important developments in General Chemistry during the last quarter of a century. **Harry C. Jones.** 326 + xii pages. Extra Crown 8vo. 8/6 net.

Some Fundamental Principles of Chemistry, Old and New. **E. A. Letts, D.Sc.,** Professor of Chemistry in the Queen's University, Belfast. Demy 8vo. 223 + xiii pages. 44 Illustrations. 8/- net.

An Introduction to the Study of Physical Metallurgy. **Walter Rosenhain, B.A., D.Sc., F.R.S.** 350 + xvi pages. 140 Illustrations. Demy 8vo. 12/6 net. ([See p. 17.](#))

Theory of Measurements. A manual for students in physics designed for use either as a text-book or as a laboratory guide. **James S. Stevens,** Professor of Physics in the University of Maine. Extra Crown 8vo. 10/- net.

Electric Arc Phenomena. **E. Rasch.** Translated by **K. Tornberg.**
Demy 8vo. 52 Illustrations. 18/- net.

CONTENTS.—General Outline of Electric Arc Phenomena. Typical Carbon Arc. Electrode Materials and their Physical Properties. Theory of Electric Discharges through Gases. Spark Discharges. Empirical Results. Voltage and Current Conditions in the Arc. Distribution of Energy in the Arc. Efficiency of Commercial Types of Arc.

Physical Laboratory Experiments for Engineering Students: Mechanics, Sound, Heat, and Light.
Samuel Sheldon, Ph.D., D.Sc., and **Erich Hausmann, F.E., Sc.D.**
Demy 8vo. 134 pages. 40 Illustrations. 7/6 net.

Radio-Active Transformations. **Ernest Rutherford, F.R.S.**
Demy 8vo. 31/6 net.

The Theory of Ionization of Gases by Collision. **John S. Townsend, M.A., F.R.S.,** Wykeham Professor of Physics in the University of Oxford. Crown 8vo. 4/6 net.

CONTENTS.—Ionization by Negative Ions. Ionization by Positive Ions. Sparking Potentials in a Uniform Electric Field. Theory of Electric Discharges in Fields of Force which are not Uniform.

The Electrical Nature of Matter and Radioactivity. **H. C. Jones,** Professor of Physical Chemistry in the Johns-Hopkins University. 3rd Edn. Revised and largely Rewritten. Demy 8vo. 220 pages. 8/6 net.

The Corpuscular Theory of Matter. **Sir J. J. Thomson, M.A., F.R.S., D.Sc., LL.D., Ph.D.,** Professor of Experimental Physics, Cambridge, and Professor of Natural Philosophy at the Royal Institution, London. Demy 8vo. 7/6 net.

Radio-Activity and Geology. **John Joly, M.A., Sc.D., F.R.S.,** Professor of Geology and Mineralogy in the University of Dublin. Crown 8vo. 10/- net.

A Text-Book of Physics. **H. E. Hurst, B.A., B.Sc.,** Hertford College, Oxford, late Demonstrator in Physics in the University

Museum, Oxford, and **R. T. Lattey, M.A.**, Royal Naval College, Dartmouth, late Demonstrator in Physics in the University Museum, Oxford. Many Illustrations and Diagrams. Demy 8vo. 12/6 net.

Also published in three volumes. Each sold separately.

Part I. DYNAMICS AND HEAT. 4/- net.

Part II. LIGHT AND SOUND. 4/- net.

Part III. MAGNETISM AND ELECTRICITY. 5/- net.

Electricity and Matter. **Sir J. J. Thomson, M.A., F.R.S., D.Sc., LL.D., Ph.D.** Crown 8vo. 9/- net.

Exercises in Physical Chemistry. **Dr. W. A. Roth**, Professor of Chemistry in the University of Greifswald. Authorised Translation by **A. T. Cameron, M.A., B.Sc.**, of the University of Manitoba. Demy 8vo. 6/- net.

Practical Physics. **Franklin, Crawford and Macnutt.**

Vol. I. PRECISE MEASUREMENTS—MECHANICS AND HEAT. 9/- net.

Vol. II. ELEMENTARY AND ADVANCED MEASUREMENTS IN ELECTRICITY AND MAGNETISM. 9/- net.

Vol. III. PHOTOMETRY. EXPERIMENTS IN LIGHT AND SOUND. 7/- net.

Atoms. **Jean Perrin.** Translated by **D. Ll. Hammick.** Demy 8vo. 16 Illustrations. 9/- net.

CONTENTS.—Chemistry and the Atomic Theory. Molecular Agitation. The Brownian Movement—Emulsions. The Laws of the Brownian Movement. Fluctuations. Light and Quanta. The Atom of Electricity. The Genesis and Destruction of Atoms. Index.

The Universe and the Atom. The Ether Constitution, Creation and Structure of Atoms, Gravitation, and Electricity, Kinetically explained. **Marion Erwin, C.E.** Demy 8vo. 8/6 net.

CONTENTS.—Part I., First Principles. Part II., The Pan Cycle Hypothesis: Invisible Composition Light Waves, the Warp and Woof of the Ether Structure and of All Things Material.

The Profession of Chemistry. **Richard B. Pilcher**, Registrar and Secretary of the Institute of Chemistry of Great Britain and Ireland. Crown 8vo. 6/6 net.

CONTENTS.—Synopsis of Training and Possible Careers for a Chemist. Preliminary Education. Pharmacists and Chemists. Professional Training. Prospects and Conditions of Practice. Professional Organisation. Public Analysts and Official Agricultural Analysts. Professional Procedure. Industrial Chemistry. Chemistry and the State. Teaching. Women in Professional Chemistry. Chemists in War.

What Industry Owes to Chemical Science. **Richard B. Pilcher** and **Frank Butler-Jones, B.A.** With an Introduction by **Sir George Beilby, LL.D., F.R.S.** Crown 8vo. 166 pages. 4/- net.

The authors show how science has advanced the methods and processes of production and laid the foundation for the establishment of new manufactures. Many important industries are surveyed with the particular object of informing students of chemistry and engineering of the vast field open to them in their professional careers.

Van Nostrand's Chemical Annual. A Handbook of Useful Data for Analytical, Manufacturing, and Investigating Chemists, and Chemical Students. Edited by **John C. Olsen, M.A., Ph.D.**, Professor of Analytical Chemistry, Polytechnic Institute, Brooklyn; with the co-operation of Eminent Chemists. Nearly 100 Tables. Crown 8vo. 18/6 net.

Principles of Microscopy. Being an Introduction to Work with the Microscope. **Sir A. E. Wright, M.D., F.R.S., D.Sc., Dublin (Honoris Causa), F.R.C.S.I. (Hon.).** With many Illustrations and Coloured Plates. Super Royal 8vo. 21/- net.

Ozone. (A Treatise of Electro-Chemistry. Edited by **Bertram Blount, F.I.C.) S. Rideal, D.Sc., F.I.C., F.C.S.** Demy 8vo. 12/- net. ([See p. 39.](#))

Metallurgy.

Methods in Metallurgical Analysis. **Charles H. White,**
Assistant Professor in Mining and Metallurgy in Harvard University.
Demy 8vo. 106 Illustrations. 21/- net.

CONTENTS.—Definition of the Subject. Sampling. Necessity for Correct Sampling. The Operations of Analysis, Gravimetric, Volumetric Analysis. Calorimetry. Methods of Analysis in the Metallurgy of Iron and Steel. Moisture. Hygroscopic Water. Combined Water. Loss of Ignition. Iron in Ores. Silica, Sulphur, Phosphorus, Alumina, Manganese, Lime, Magnesia and Titanium in Ore. Analysis of—Iron and Steel, Iron Slags, Limestone. Methods of Analysis in the Metallurgy of Copper, Lead, etc. Copper, Lead, Zinc and Arsenic in Ore. Analysis of—Copper Matte, Chilled Blast Furnace Slags, Reverberatory Slags, Briquettes and other Copper-bearing Products, Copper Bullion, Alloys. Methods of Analysis in the Production of the Precious Metals. Analysis of Fluxes. Analysis of Fuels. Analysis of Clay. Methods for the Determination of Some of the Minor Metals. Methods for the Determination of some of the Rarer Metals. Testing of Lubricating Oils. Examination of Boiler Water. Detection of the Metals. Tables. General References.

An Introduction to the Study of Physical Metallurgy.
Walter Rosenhain, B.A., D.Sc., F.R.S. 140 Illustrations. Demy 8vo.
12/6 net.

CONTENTS.—The Structure and Constitution of Metals and Alloys. The Microscopic Examination of Metals. The Metallurgical Microscope. The Micro-Structure of Pure Metals and Alloys. The Thermal Study of Alloys. The Constitutional Diagram and the Physical Properties of Alloys. Typical Alloy Systems. The Iron-Carbon System. The Properties of Metals as Related to their Structure and Constitution. The Mechanical Testing of Metals. The Mechanical Testing of Metals (*contd.*). The Effect of Strain on the Structure of Metals. The Thermal Treatment of Metals. The Mechanical Treatment of Metals, including Casting. Defects and Failures in Metals and Alloys.

The Electro-Metallurgy of Steel. **Colin Gow, A.R.S.M., B.Sc.**
Edited by **Bertram Blount.** Demy 8vo. Illustrated. About 24/- net.

Mathematics.

The Theory of Proportion. **M. J. M. Hill, M.A., LL.D., Sc.D., F.R.S.** Demy 8vo. 15 Figures. 6/- net.

CONTENTS.—Magnitudes of the same Kind. Propositions relating to Magnitudes and their Multiples. The Relations between Multiples of the same Magnitude. Commensurable Magnitudes. Magnitudes of the same Kind which are not Multiples of the same Magnitude. Incommensurable Magnitudes. Extension of the Idea of Number. On the Ratios of Magnitudes which have no Common Measure. Properties of Equal Ratios. First Group of Propositions. Second Group of Propositions. Third Group of Propositions—Geometrical Applications of Stolz's Theorem. Further Remarks on Irrational Numbers. The Existence of the Fourth Proportional. Commentary on the Fifth Book of Euclid's Elements.

Elementary Plane Trigonometry. **H. E. Piggott, M.A.** Second Master and Head of Mathematical Department, Royal Naval College, Dartmouth. Late Director of Naval Studies at Cambridge, Formerly Scholar of Clare College, Cambridge. Crown 8vo. 7/6 net.

CONTENTS.—The Tangent of an Acute Angle. The Sine and Cosine of an Acute Angle. Miscellaneous Application of the Sine, Cosine and Tangent. The Co-tangent, Secant, and Co-secant of an Acute Angle. Exercises of the Six Trigonometrical Ratios. Miscellaneous Revision Examples. Solution of Triangles. Miscellaneous Problems. Areas of Triangles, etc. Circular Measure. The Sphere. Trigonometrical Ratios of Angles Greater than 180° . Directed Lines. Compound Angles. Graphs of Periodic Functions. Appendix I. Traverse Table. Appendix II: Haversines. Miscellaneous Examples. Answers to Examples. Index.

Integration by Trigonometric and Imaginary Substitution. **Charles O. Gunther**, with an Introduction by **J. Burkett Webb, C.E.** Demy 8vo. 5/- net.

Quaternions, as the Result of Algebraic Operations. **Arthur Latham Baker, Ph.D.** Extra Crown 8vo. 6/- net.

The Calculus and its Applications. A Practical Treatise for Beginners, especially Engineering Students. With over 400 examples, many of them fully worked out. **Robert Gordon Blaine, M.E., A.M.Inst.C.E.**, formerly Lecturer at the City Guilds' Technical College. Crown 8vo. 5/- net.

An Elementary Treatise on the Calculus. **Franklin, MacNutt** and **Charles**. 16/6 net.

Elementary Statics. **Franklin** and **MacNutt**. 4/- net.

Mathematics for the Practical Man. A short work explaining all the elements of Algebra, Geometry, Trigonometry, Logarithms, Co-ordinate Geometry, and Calculus. **George Howe, M.E.** Crown 8vo. 7/- net.

The Practical Mechanic's Handbook. **F. E. Smith**. 8/- net.

CONTENTS.—Arithmetic. Arithmetical Signs and Characters and Explanation of Solving Formula. Mensuration. How the Dimensions, Measurements, and Weights of Different Shaped Vessels are Found. The Primary or Simple Machines. Strength of Materials and Questions Relating to Stress.

Manufacture and Industries.

Application of Physico-Chemical Theory to Technical Processes and Manufacturing Methods. **Dr. R.**

Kremann. Translated into English and Edited by **H. E. Potts, M.Sc.**
Demy 8vo. 9/- net.

CONTENTS.—The Two Fundamental Laws of the Mechanical Theory of Heat. Reaction-Velocity and Catalytes. Other Special Applications of the Law of Mass Action. The Influence of Temperature on the Equilibrium-Constant. Dissociation Pressure. Application of the Phase Rule. The Application of the Phase Rule to Solid-Liquid Systems. Transformation Phenomena in Hydraulic Binding Agents. Other Applications of the Phase Rule. The Distribution Law. Adsorption Compounds. Reciprocal Pairs of Salts.

Chemistry of the Oil Industries. **J. E. Southcombe, M.Sc.,**

Lecturer on Oils and Fats, Royal Salford Technical Institute, etc. 9/- net.

CONTENTS.—Preface. Introductory Organic Chemistry. Mineral Oils. Petroleum and Shale. Mineral Oil Refining. Natural Sources and Methods of Preparation of the Saponifiable Oils and Fats. Impurities occurring in Crude Oils and Fats and the Technical Methods of Removing them. Composition and Properties of the Saponifiable Oils and Fats in General. Composition and Properties of the Individual Oils and Fats of Commercial Importance. The Natural Waxes, their Composition and Properties. Analytical Methods. Industrial Applications of Fats and Oils. Burning Oils. Edible Oils and Margarines. Polymerised, Boiled and Blown Oils. Turkey-Red Oils. Saponification of Fats and Oils on a Technical Scale. The Distillation of Fatty Acids. Oleines and Stearines. Candle Manufacture. Soap-Making. Glycerine. Conclusion: Scientific and Technical Research on Problems in the Oil and Related Industries. Literature.

Linseed Oil and other Seed Oils. An Industrial Manual.

William D. Ennis, M.E., M.Am.Soc.M.E., Professor of Mechanical Engineering, Polytechnic Institute of Brooklyn. Medium 8vo. Illustrated. 37/6 net.

The Production and Treatment of Vegetable Oils.

Including Chapters on the Refining of Oils, Hydrogenation of Oils, Generation of Hydrogen, Soap Making, Recovery and Refining of Glycerine, and the Splitting of Oils. **T. W. Chalmers, B.Sc.,**

A.M.I.Mech.E. Crown 4to. 164 pages. 9 Folding Plates. 95 Illustrations. 21/- net.

CONTENTS.—Introductory. Principal Vegetable Oils. Preparatory Machinery for Copra and Linseed, Palm Fruits and Palm Kernels, Cotton Seed and Castor Seed. Special Forms of Reduction Machinery. Meal Kettles, Receiving Pans and Moulding Machines. Oil Presses. Anglo-American and Cage Type. General Arrangement of Oil Mills. Extraction of Oils by Chemical Solvents. Refining of Oils. Hydro-generation or Hardening of Oils. Generation of Hydrogen for Oil Hardening Purposes. Manufacture of Soap. Glycerine Recovery and Refining and the Splitting of Oils.

Cotton Seed Products. A Manual of the Treatment of Cotton Seed for its Products and their Utilization in the Arts. **Leebert Lloyd Lamborn, M.Am.C.S., M.S.C.I.** With 79 Illustrations and a Map. Demy 8vo. 31/6 net.

Paper Making and its Machinery. **T. W. Chalmers.** Imperial 8vo. 200 pages. Illustrated. 26/- net.

CONTENTS.—Introduction. Cutting, Cleaning and Boiling. Washing, Breaking and Bleaching. Purifying and Pulping. Beating. Refining. The Fourdrinier Machine. Fourdrinier Driving Arrangements. Details of the Fourdrinier. Pulp Strainers. Tub Sizing. Calendering, Cutting and Winding. Wood Pulp. The Coating of Art Paper. The Finishing of Coated Art Paper. The Coating of Photographic Paper.

The Manufacture of Paper. **R. W. Sindall, F.C.S.** 2nd Edn. Extra Crown 8vo. 8/6 net.

CONTENTS.—Preface. List of Illustrations. Historical Notice, Cellulose and Paper-Making Fibres. The Manufacture of Paper from Rags, Esparto and Straw. Wood Pulp and Wood Pulp Papers. Brown Papers and Boards. Special kinds of Paper. Chemicals used in Paper-making. The Process of "Beating." The Dyeing and Colouring of Paper Pulp. Paper Mill Machinery. The Deterioration of Paper. Bibliography. Index.

Wood Pulp. **Charles F. Cross, B.Sc., F.I.C.; E. J. Bevan, F.I.C., and R. W. Sindall, F.C.S.** New and Revised Edition. 8/6 net.

CONTENTS.—The Structural Elements of Wood. Cellulose as a Chemical. Sources of Supply. Mechanical Wood Pulp. Chemical Wood Pulp. The Bleaching of Wood Pulp. News and Printings. Wood Pulp Boards. Utilisation of Wood Waste. Testing of Wood Pulp for Moisture. Wood Pulp and the Textile Industries. Bibliography. Index.

Commercial Paints and Painting. A Handbook for Architects, Engineers, Property Owners, Painters and Decorators, etc. **Arthur Seymour Jennings.** 224 + xii pages. 47 Illustrations. Extra Crown 8vo. 7/6 net.

Glass Manufacture. **Walter Rosenhain, D.Sc., F.R.S.**
Superintendent of the Department of Metallurgy in the National
Physical Laboratory. 2nd and Revised Edition. 12/6 net.

CONTENTS.—Physical and Chemical Properties of Glass. The Physical Properties of Glass. The Raw Materials of Glass Manufacture. Refractories. Furnaces. The Process of Fusion. Processes Used in the working of Glass. Bottle Glass. Blown and Pressed Glass. Rolled or Plate Glass. Sheet and Crown Glass. Coloured Glasses. Optical Glass. Miscellaneous Products.

Decorative Glass Processes. **Arthur Louis Duthie.** Extra
Crown 8vo. 8/6 net.

CONTENTS.—Introduction. Various Kinds of Glass in Use. Their Characteristics, Comparative Price, etc. Leaded Lights. Stained Glass. Embossed Glass. Brilliant Cutting and Beveling. Sand-Blast and Crystalline Glass. Gilding. Silvering and Mosaic. Proprietary Processes. Patents. Glossary.

The Manufacture of Leather. **Hugh Garner Bennett, M.Sc.,
F.C.S.,** Member of the International Association of Leather Trade
Chemists. Demy 8vo. 18/- net.

CONTENTS.—Historical and Introductory Outline. The Nature of Skin. Fermentation. Hides and Skins. Water. Soaking. Unhairing. Deliming. The Tannins. The Vegetable Tanning Materials. The Analysis of Tanning Materials. The Preparation of the Tanning Liquors. The Principles of Vegetable Tanning. The Tannage of Sole Leather. The Tannage of Belting, Harness and Upper Leather, etc. The Tannage of Moroccos and Light Leathers. The Tannage of Chrome Leather. Alum and Combination Tannages. Fat, Oil and Aldehyde Tannages. The Drying of Leather. The Finishing of Sole Leather. The Currying and Finishing of Dressing Leather. Leather Dyeing. The Finishing of Light Leathers. The Finishing of Chrome Leather. The Finishing of the Alum- and Combination-Tanned Leathers. The Finishing of Fat- and Oil-Tanned Leathers. Japanned and Enamelled Leathers. The Dressing of Wool Rugs. The Analysis of Leather. Index.

*Mineral and Aerated Waters, and the Machinery for
their Manufacture.* **C. Answorth Mitchell, B.A. (Oxon.),
F.I.C.** Demy 8vo. 9/- net.

The Chemistry and Technology of Printing Inks.
Norman Underwood and Thomas V. Sullivan. Medium 8vo. 12/6
net.

Materials for Permanent Painting. A Manual for
Manufacturers, Art Dealers, Artists and Collectors, explaining the
Composition of the Materials used in Painting. **M. Toch, F.C.S.**
Frontispiece in Colour and other Illustrations. Crown 8vo. 8/- net.

Materials Used in Sizing. **W. F. A. Ermen.** 6/- net.

CONTENTS.—The Starches and other Agglutinants. Weighing Materials. Softening Ingredients. Antiseptics. Analysis of Sized Warps and Cloth. The Preparation of Normal Volumetric Solutions. Tables.

The Nitrocellulose Industry. **Edward Chauncey Warden,**
Ph.C., M.A., F.C.S. 2 Volumes ([See p. 44.](#)) 50/-.

Oils, Gums, Waxes and Allied Substances. **F. S. Hyde,**
Ph.B. Demy 8vo. 12/6 net.

India-Rubber and its Manufacture. With Chapters on Gutta-
Percha and Balata. **H. L. Terry, F.I.C., A.I.M.M.** Extra Crown 8vo.
8/6 net.

The Chemistry of the Rubber Industry. **Harold E. Potts,**
M.Sc. ([See p. 43.](#))

Solvents, Oils, Gums, Waxes, and Allied Substances.
F. S. Hyde, Ph.B. Demy 8vo. 12/- net.

Glues and Gelatine. A Practical Treatise on the Methods of Testing
and Use. **R. Livingston Fernbach.** Demy 8vo. 14/- net.

*Agglutinants and Adhesives of all Kinds for all
Purposes.* **H. C. Standage.** Demy 8vo. 7/6 net.

*The Chemistry of Dyeing and Bleaching of Vegetable
Fibrous Material.* **Julius Hübner, M.Sc.Tech., F.I.C.** ([See p. 44.](#))

Cork: Its Origin and Industrial Uses. **Gilbert E. Stecher.**
Crown 8vo. 6/- net.

Wool. **Frank Ormerod.** Demy 8vo. Illustrated. 6/6 net. ([See p. 3.](#))

Cotton. **George Bigwood.** Demy 8vo. Illustrated. 6/6 net. ([See p. 3.](#))

History of the Frozen Meat Trade. **J. C. Critchell** and **J. Raymond.** Demy 8vo. 12/6 net.

The Potter's Craft. A Practical Guide for the Studio and Workshop.
Charles F. Binns, late Superintendent Royal Porcelain Works,
Worcester. With 42 Illustrations. Crown 8vo. 7/6 net.

Arts and Crafts.

Photography: Its Principles and Applications. **Alfred Watkins, F.R.P.S.**, Past President Photographic Convention. Demy 8vo. 340 pages. 100 Illustrations. 12/6 net. 2nd Revised Edition.

CONTENTS.—First Principles. Lenses. Exposure Influences. Practical Exposure. Development Influences. Practical Development. Cameras and Dark Room. Orthochromatic Photography. Printing Processes. Hand camera Work. Enlarging and Slide Making. Colour Photography. General Applications. Record Applications. Science Applications. Plate Speed Testing. Process Work. Pinhole Photography.

Simple Jewellery. A Practical Handbook with Certain Elementary Methods of Design and Construction. Written for the use of Craftsmen, Designers, Students and Teachers. **R. Ll. B. Rathbone.** 2nd Edn. In preparation.

Precious Stones. **W. Goodchild, M.B., B.Ch.** With a Chapter on Artificial Stones by **Robert Dykes.** With 42 Illustrations. Extra Crown 8vo. 7/6 net.

CONTENTS.—Introduction. Modes of Origin. Physical Properties of Gem Stones. Cutting of Gems. Imitation Gems and the Artificial Production of Precious Stones. Diamond. Fluor Spar. Opal. Corundum. Spinel and Chrysoberyl. Calcite. Labradorite. Beryl. Garnet. Olivine. Sphene. Apatite. Jet. Glossary.

Business and Management.

Advertising: A Study of a Modern Business Power. **G. W. Goodall, B.Sc.** With an Introduction by **Sidney Webb, LL.B.** Paper, 1/6.

Storing: Its Economic Aspects and Proper Methods. **H. B. Twyford.** Author of "Purchasing." Large Demy 8vo. 21/6 net.

Purchasing: Its Economic Aspects and Proper Methods. **H. B. Twyford,** of the Otis Elevator Company. 112 Charts, Diagrams and Forms. Large Demy 8vo. 252 pages. 26/- net.

Seasonal Trades. **Arnold Freeman, B.A.,** and **Sidney Webb.** With a Bibliography of each Trade. 7/6 net.

Commerce and Industry. Edited by **William Page, F.S.A.** With a Preface by **Sir William Ashley, M.A., Ph.D.** Each volume is sold separately.

Vol. I. A HISTORICAL REVIEW OF THE ECONOMIC CONDITIONS OF THE BRITISH EMPIRE FROM THE PEACE OF PARIS IN 1815 TO THE DECLARATION OF WAR IN 1914, BASED ON PARLIAMENTARY DEBATES. 508 Pages and 7 Maps. 32/- net.

CONTENTS.—Editorial Note. Preface. Introduction. Effects of War (1815 to 1820). Commercial Reform (1820 to 1830). The Reform Parliament (1830 to 1841). Repeal of the Corn Laws (1841 to 1852). War and Finance (1852 to 1859). Free Trade (1859 to 1868). Retrenchment and Reform (1868 to 1880). Organization (1880 to 1892). Foreign Competition (1892 to 1900). The Movement towards Tariff Reform (1900 to 1910). Unrest (1910 to 1914). The Cabinet and Parliament. Ministries (1812 to 1912). A Chronicle of the British Empire Beyond the Seas. Index. Maps.

Vol. II. TABLES OF STATISTICS FOR THE BRITISH EMPIRE FROM 1815. 260 pages. 24/- net.

CONTENTS.—Populations. Revenue and Expenditure. National Debt of the United Kingdom. Public Debt of India and the Principal Colonies. Local Taxation. Expenditure for Education. Imports and Exports of the United Kingdom. Imports and Exports of India and the Principal Colonies: Imports of Bullion and Specie. Shipping. Railways. Light Railways and Tramways. Production, the United

Kingdom. Production, The British Empire. Wages. Prices. Bank Discount. Savings Banks in the United Kingdom. Textile Factories. Registered Companies. Gas Undertakings. Electrical Undertakings.

Primer of Scientific Management. **F. B. Gilbreth, M.Am.Soc.M.E.**, with an Introduction by **Louis D. Brandeis.** 9/- net.

Motion Study. A Method for Increasing the Efficiency of the Workman. **F. B. Gilbreth, M.Am.Soc.M.E.** 9/6 net.

Useful Handbooks and Tables, etc.

The Practical Mechanic's Handbook. **F. E. Smith.** 8/- net.
([See p. 51.](#))

Materials Used in Sizing. **W. F. A. Ermen.** ([See p. 54.](#))

Reference Book for Statical Calculations. **Francis Ruff.**
Crown 8vo. 160 Illustrations. Charts and Diagrams. 4/- net.

The New Steam Tables. Calculated from Professor Callendar's
Researches. **Professor C. A. M. Smith, M.Sc., and A. G. Warren,**
S.Sc. 4/6 net.

*Handbook for the Care and Operation of Naval
Machinery.* **Comr. H. C. Dinger,** U.S. Navy. 436 pages. 150
Illustrations. Pocket size. 21/- net. 3rd Edn. Revised.

CONTENTS.—Operation of Naval Machinery. Care and Overhaul of Main Plant. Fittings and
Auxiliaries. Care and Preservation of Subdivision of Hull. Special Auxiliary Engines. Spare Parts.
Tests of Machinery and Piping.

The Law affecting Engineers. **W. Valentine Ball, M.A.**
(Cantab.), Barrister-at-Law, Joint Editor of "Emden's Building
Contracts." Demy 8vo. 10/6 net.

A concise statement of the powers and duties of an engineer, as between employer and contractor,
as arbitrator and as expert witness, together with an outline of the law relating to engineering
contracts, and an appendix of forms of contract, with explanatory notes.

Lettering. For Draughtsmen, Engineers and Students. **C. W. Reinhardt.**
14th Edn. Revised and Enlarged. 10/- net.

CONTENTS.—Inclined and Upright Lettering: Lower-case Letters. Capital Letters, Numerals.
Extending Form. Freehand Lettering Applied to Working Drawings. Various Freehand Alphabets.
Greek Alphabet. Lettering of Titles. Practice Sheets, Round Writing. Lettering for Photo-
reproduction.

Natural History, Botany, Nature Study, etc.

Extinct Animals. **Sir E. Ray Lankester, F.R.S.** With a Portrait of the Author and 218 other Illustrations. New and Revised Edition in preparation. Extra Crown 8vo.

Economic Zoology and Entomology. **Vernon L. Kellogg and C. W. Doane.** Crown 8vo. 6/6 net.

The authors are well known as clear and precise exponents of their subject. In his text-book they have produced an introduction both to general zoology as well as to that specific phase of it called "economic."

The first chapters introduce the reader to general facts of animal structure and life and are arranged on the basis of accepted pedagogic principles. The latter chapters, arranged on a basis of animal classification proceeding from the samples to the nine highly developed groups, include not only general facts pertaining to the groups treated, but introduce and give general attention to the economic relations of the various members of the groups. Finally, there is presented a form of encyclopædic treatment of a wide range of facts wholly economic in aspect.

Distribution and Origin of Life in America. **R. F. Scharff, B.Sc., Ph.D., F.L.S.** Fully Illustrated. Large Crown 8vo. 10/6 net.

European Animals: Their Geological History and their Geographical Distribution. **R. F. Scharff, B.Sc., Ph.D., F.L.S.** Fully Illustrated. Demy 8vo. 7/6 net.

Plant Physiology and Ecology. **Frederic Edward Clements, Ph.D.,** Professor of Botany in the University of Minnesota. With 125 Illustrations. Demy 8vo. 12/- net.

The Living Plant. A Description and Interpretation of its Functions and Structure. **William F. Ganong, Ph.D.** Fully Illustrated. 465 + xii pages. 15/- net.

Plants and their Uses. An Introduction to Botany. **F. L. Sargent.** 612 pages. 384 Illustrations. Crown 8vo. 7/6 net.

The Flower and the Bee. Plant Life and Pollination. **John H. Lovell.** Extra Crown 8vo. Profusely illustrated by Photographs taken by the Author. 10/6 net.

A popular exposition, suitable to amateur gardeners, bee-keepers and lovers of nature generally.

Life Histories of Northern Animals. An Account of the Mammals of Manitoba. **Ernest Thompson Seton,** Naturalist to the Government of Manitoba. In 2 volumes. Large 8vo. Over 600 pages each. With 70 Maps and 600 Drawings by the Author. 73/6 net.

Influences of Geographic Environment. **E. C. Semple,** Author of "American History and its Geographic Conditions." Medium 8vo. 700 pages. 21/- net.

Outlines of Evolutionary Biology. **Arthur Dendy, D.Sc., F.R.S.,** Professor of Zoology in the University of London (King's College); Zoological Secretary of the Linnean Society of London; Honorary Member of the New Zealand Institute; formerly Professor of Biology in the Canterbury College (University of New Zealand), and Professor of Zoology in the South African College, Cape Town. 2nd Edn. Enlarged. 4th Impression, with a Glossary. Fully Illustrated. 15/- net.

Agriculture, Farming and Forestry.

Agriculture and Irrigation. **Kinsley D. Doyle.** Demy 8vo. 288 pp. 19/- net.

CONTENTS.—The Soil. Fertilizers. Irrigation. Wells and Pumping. Dry Farming and Economics. Pastures, Natural and Artificial. Notes on Forests, Cattle, Pigs and Carobs. Runs and Farms for Sheep and Ostriches. Cereals. Apples. Asparagus. Onion and Potatoes. Citrus Orchards. Grapes, Walnuts and Pistachio Nuts. Some Special Industries. Some Tropical Fruit. Tropical Plantations. Planting and Sowing for Oil. Tropical Agriculture.

Dairy Laboratory Guide. **C. W. Melick, B.S.A., M.S.** Crown 8vo. With 52 Illustrations.

Soils and Manures. **J. Alan Murray, B.Sc.** Extra Crown 8vo. 8/6 net.

CONTENTS.—Preface. Introductory. The Origin of Soils. Physical Properties of Soils. Chemistry of Soils. Biology of Soils. Fertility. Principles of manuring. Phosphatic Manures. Phospho-Nitrogenous Manures. Nitrogenous Manures. Potash Manures. Compound and Miscellaneous Manures. Potash Manures. Compound and Miscellaneous Manures. General Manures. Farmyard Manure. Valuation of Manures. Composition and Manurial Value of Various Farm Foods. Index.

The First Book of Farming. **Charles L. Goodrich.** With 87 Illustrations. Crown 8vo. 7/6 net.

CONTENTS.—General Principles Underlying Plant Culture. Introduction to Plants. Roots. Soils. Relation of Soils to Water. Forms of Soil Water. Loss of Soil Water. Soil Temperature. Plant Food in the Soil. Seeds. Planting. Spading and Ploughing. Harrowing and Rolling. Leaves. Stems. Flowers. Soil Fertility as Affected by Farm Operations and Farm Practices. Fertile Soils. Soil Water. After-Cultivation of Crops. Farm Manures. Commercial Fertilizers. Rotation of Crops. Farm Drainage. Glossary.

Commercial Egg Farming. **S. G. Hanson.** Crown 8vo. Illustrated. 7th Impression. 1/9 net.

A practical handbook based on the experience of the successful poultry farmer. Deals with breeding, incubating, brooder-house management, colony houses, laying houses, and the co-operative marketing of eggs. Tables showing outlay required for establishing farms of various sizes are included.

Utility Ducks and Geese. Their successful Rearing, Management, and Marketing. **J. W. Hurst**, Poultry Correspondent of "The Field." Crown 8vo. Illustrated. 2/6 net.

Insect Pests and Plant Diseases in the Vegetable and Fruit Garden. **F. Martin Duncan, F.R.M.S., F.R.P.S.**, Author of "Our Insect Friends and Foes," "Wonders of Insect Life," "Cassell's Natural History," etc. Crown 8vo. Illustrated. 3/6 net.

The object of this little book is to give in as brief and concise a form as possible a simple account of those Insect Pests and Plant Diseases with which the Allotment-holder and Gardener will be most likely to meet in the course of their labours. Keeping in mind that this work is intended for the help of the Amateur, care has been taken so far as possible to avoid the use of technical terms, and to explain their meaning when it was considered necessary.

Farm Animals: How to Breed, Feed, Care For and Use Them. **E. V. Willcox, Ph.D., M.A.**, U.S.A. Department of Agriculture. With over 60 Full-page Illustrations. Demy 8vo. 10/- net.

Coniferous Trees. Being a Concise Description of each Species and Variety, with the most recently approved Nomenclature, list of Synonyms, and best Methods of Cultivation. **A. D. Webster**, Author of "Practical Forestry," "The Forester's Diary," etc. Demy 8vo. 320 pages. 28 Plates. 21/- net.

A complete manual on conifers, with chapters on the economic aspect, the quality and value of British grown timber, soils and situations most suited to the trees, their characteristics, means of propagation and the enemies to which they are subject.

Estate Economics. **Andrew Slater**, Gold Medallist of the Highland and Agricultural, and of the Royal Scottish Arboricultural Societies; late Land Agent to the War Department in the Southern, Northern, and Scottish Commands; and previously Land Steward to their late Majesties Queen Victoria and King Edward at Osborne. Demy 8vo. Fully Illustrated. 10/- net.

CONTENTS.—The Origin of Soils. The Drainage of Soils. Embanking of Rivers. Roads, etc.: Their construction and Maintenance. Fences and Gates, etc.: Division (1) Preliminary; (2) Hedges; (3) Walls; (4) Wooden Fences; (5) Gates, etc. Building, etc.: Division (1) Preliminary; (2) Hay Sheds; (3) Sheep Folds; (4) Motive Power. Water Supplies. Sewage Drains, etc. The Utilisation of Land: Division (1) Preliminary; (2) Manuring; (3) Cultivation; (4) Permanent Pastures; (5) Hay Meadows;

(6) Water Meadows; (7) Orchards; (8) Upland Pastures; (9) Reclamation of Land. Motor Traction. Forestry: Division (1) Preliminary; (2) The Tree and its Life History; (3) British Forest Trees; (4) The Nursery; (5) The Formation of Woodland Crops; (6) The Management of Woodland Crops; (7) The Treatment of Matured Woodland Crops; (8) The Disposal of Woodland Crops; (9) The Valuation of Woodland Crops; (10) Forest Working Plans; (11) General; (12) The Preservation of Timber.

Forests, Woods and Trees in Relation to Hygiene.

Augustine Henry, M.A., F.L.S., M.R.I.A., Professor of Forestry, Royal College of Science, Dublin. Demy 8vo. Fully Illustrated with Maps and Photographs. 18/- net.

CONTENTS.—The Influence of Forests on Climate. The Sanitary Influence of Forests. Forests as Sites for Sanatoria. Parks in Town and Municipal Forests. Trees in Towns. Afforestation of Pit Mounds. Afforestation of Water Catchment Areas. Conditions affecting the Planting of Water Catchment Areas. Trees for Water Catchment Areas and for General Planting. Water Catchment Areas in England and Wales. Water Catchment Areas in Scotland. Water Catchment Areas in Ireland. Index.

Indian Trees. An Account of Trees, Shrubs, Woody Climbers, Bamboos and Palms, Indigenous or Commonly Cultivated in the British Indian Empire. **Sir Dietrich Brandis, K.C.I.E., Ph.D.(Bonn), LL.D.(Edin.), F.R.S., F.L.S., F.R.G.S.,** and Hon. Member of the Royal Scottish Arboricultural Society, of the Society of American Foresters, and of the Pharmaceutical Society of Great Britain. Assisted by Indian Foresters. New Edition in Preparation. 40/- net.

Law, Patents, etc.

Building in London. A Treatise on the Law and Practice affecting the Erection and Maintenance of Buildings in the Metropolis. **Horace Cubitt, A.R.I.B.A.**, etc. Illustrated, with Diagrams. 31/6 net.

Industrial Accidents and their Compensation. **G. L. Campbell, B.S.** Crown 8vo. 12/- net.

The Law affecting Engineers. **W. Valentine Ball, M.A. (Cantab.)**, Barrister-at-Law, Joint Editor of "Emden's Building Contracts." Demy 8vo. 10/6 net.

Henslowe's Motor Dictionary. Compiled by **Leonard P. Henslowe**. English-French, French-English. Contains British and Metric Units of Measurement.

Foreign and Colonial Patent Laws. **W. Cranston Fairweather**, Chartered Patent Agent. Demy 8vo. 10/6 net.

Patents, Designs and Trade Marks. The Law and Commercial Usage. **Kenneth R. Swan, B.A.(Oxon.)**, of the Inner Temple, Barrister-at-law. Extra Crown 8vo. 7/6 net.

The Arbitration Clause in Engineering and Building Contracts. **E. J. Rimmer (Barrister-at-Law), M.Inst.C.E.**, etc. Crown 8vo. 2/- net.

Miscellaneous.

Dictionary of British Scientific Instruments. Prepared by the **British Optical Instrument Manufacturers' Association**. Demy 8vo. Fully Illustrated. 21/- net.

The British in China, and Far Eastern Trade. **C. A. Middleton Smith, B.Sc., A.M.I.M.E.** Demy 8vo. 18/- net.

Forecasting Weather. **Sir W. N. Shaw, F.R.S., Sc.D.,** etc., Director of the Meteorological Office, London. New Edition in Preparation.

Clouds. A Descriptive Illustrated Guidebook to the Observation and Classification of Clouds. **George Aubourne Clarke, F.R.P.S., F.R.Met.Soc.** With a Preface by **Sir Napier Shaw, LL.D., Sc.D., F.R.S.,** Director of the Meteorological Office. Demy 8vo. 148 pp. Illustrated. 21/- net.

Town Planning in Madras. A Review of the Conditions and Requirements of City Improvement generally and Development in the Madras Presidency in particular. **H. V. Lanchester, F.R.I.B.A., M.T.P.I.,** etc. 4to. 124 pages Text. 66 Full-page and 13 Folding Plates. 12/6 net. ([See p. 31.](#))

Soft Soldering, Hard Soldering, and Brazing. A Practical Treatise on Tools, Materials, and Operations, for the use of Metal Workers, Plumbers, Tinnerns, Mechanics and Manufacturers. **James F. Hobart, M.E.** 203 pages. 62 Illustrations. 9/6 net.

Perspective. An Elementary Text-Book. **B. J. Lubschez, M.Am.I.A.** 88 pages. 32 Illustrations. 6/- net.

Public Health Chemical Analysis. **Robert C. Frederick,** Analytical Assistant to the Professor of Hygiene, Royal Naval College,

Greenwich, and **Aquila Forster, Ph.D., M.Sc., A.I.C.**, Research Chemist, Research Department, Royal Arsenal, Woolwich. Demy 8vo. 305 pp. Illustrated. 21/- net.

CONTENTS.—Introductory. Air. Water. Sewage, etc. Milk and Cream. Butter and Margarine. Lard. Carbohydrates. Proprietary Foods. Rice. Alcoholic Liquors. Baking Powders. Tea, Coffee, etc. Metallic Poisons in Food-Stuffs. Disinfectants. Soap. Rag Flock. Appendix. Index.

Colour and Its Applications. **M. Luckiesh**, Physicist, Nela Research Laboratory, National Lamp Works of General Electric Co. 129 Illustrations, including 4 Plates in Colour. 18/- net.

Light and Shade and Their Applications. **M. Luckiesh**. Medium 8vo. 280 pages. 135 Illustrations and Tables. 15/- net.

Natural Rock Asphalts and Bitumens. Their Geology, History, Properties and Industrial Application. **Arthur Danby**, Chemist and Public Consultant on Rock Asphalt and Bitumen. Crown 8vo. 244 + ix pages. 8/6 net.

CONTENTS.—Nomenclature and Definitions. Geology of Bitumen and Rock Asphalt. Appearance and Physical Structure. History and Ancient Use. Modern Exploitation of Rock Asphalt. Sources of Rock Asphalt and Bitumen. American Deposits of Bitumen. Extraction and Preparation of Rock Asphalt. Tests and Analyses. Physical Properties of Rock Asphalt. The Carrying Out of Rock Asphalt Work, and of Rock Asphalt Mastic Work. Macadam Roads. Other Uses of Bitumen.

Water, its Purification and Use in the Industries. **w. w. Christie, M.Am.S.M.E., M.Am.S.C.E.** 230 pages. 79 Illustrations. 12/6 net.

Time and Clocks. A Description of Ancient and Modern Methods of Measuring Time. **H. H. Cunynghame, M.A., C.B.** Crown 8vo. 3/6 net.

Structural Geology. **C. K. Leith**. 169 + viii pages. 68 Illustrations. Demy 8vo. 8/6 net.

Seasonal Trades. **Arnold Freeman, B.A., and Sidney Webb**. With a Bibliography of each trade. 7/6 net.

Industrial Engineering: Its Present Position and Post-War Outlook. **F. W. Lanchester, M.Inst.C.E.** Crown 8vo. Paper, 1/- net.

The Limitations of Science. **Louis T. More, Ph.D.**, Professor in the University of Cincinnati. Crown 8vo. 6/6 net.

CONTENTS.—Science as a Symbol and a Law. The Metaphysical Tendencies of Modern Physics. Descartes and his Influence on Recent Science. The Scientific Method. The Classical and the New Mechanics. Scepticism and Idolatry in Science. Science as the Arbiter of Ethics. Index.

The Seven Follies of Science. **John Phin.** A New and Enlarged Edition. Demy 8vo. 15/- net.

A popular account of the most famous scientific impossibilities, and the attempts which have been made to solve them. To which is added a small budget of interesting paradoxes, illusions and marvels.

The Claims of Labour and of Capital. **W. R. Cooper.** Crown 8vo. 2/6 net.

Terms of Industrial Peace. **Alex. Ramsay.** Crown 8vo. 3/6 net.

A vigorous and well-informed survey of the relations existing between Capital and Labour. The author emphasises the necessity of industrial peace in view of the need for greater productive power in this country, and outlines a scheme for the avoidance and settlement of disputes.

British Railways. A Financial and Commercial Study. **W. R. Lawson**, Chairman of the Railway Shareholders' Association. 320 + xxxii pages. Extra Crown 8vo. 6/- net.

The Sea Fisheries. **J. T. Jenkins, D.Sc., Ph.D.** Royal 8vo. 24/- net. Very fully Illustrated.

CONTENTS.—Statistical Methods. The Collection of Statistics. Methods of Fishing. The Trawling Grounds. Consideration of Individual Fish. The Rise of the Herring Fisheries. The Development of Steam Trawling. Legislation and the Sea Fisheries. The Inshore Fisheries. Public Fisheries for Shellfish. The Education of Fishermen. Science and the Sea Fisheries. State Aid and Fishery Research. Foreign and Colonial Fisheries and their Administration.

A Text-Book of Oceanography. **J. Travis Jenkins**, Author of "The Sea Fisheries." Demy 8vo. 256 pp. Illustrated. 15/- net.

CONTENTS.—Introduction. Extent of the Ocean. Classification of the Oceans and Seas. Sea Level. Hydrographical Curve of the Earth's Surface. General Features of the Ocean Bottom. Ocean Deep. Oceanic Deposits and Bottom Fauna. The Temperature of the Sea. The Properties of Sea-Water. Sea-Ice. Icebergs. Atmospheric Gases in Sea-Water. Sea-Water as a Food Solution for Plants. Salinity. Waves. The Tides. Ocean Currents. Appendices.

The Curves of Life. Being an Account of Spiral Formations and their Application to Growth in Nature, to Science, and to Art. **Sir Theodore Andrea Cook, M.A., F.S.A.** 415 Illustrations and 11 Plates. Royal 8vo. 470 + xxxii pages. 12/6 net.

The World's Meat Future. An Account of the Live Stock Position and Meat Prospects of all Leading Stock Countries of the World with Full Lists of Freezing Works. **A. W. Pearse, F.R.G.S.,** Editor of "The Pastoral Review." With an Introduction by **Sir Owen Cox, K.B.E.** Demy 8vo. 352 pages. 291 Illustrations. 2nd Ed. Price 21/- net.

CONTENTS.—The World's Meat Future. Summary of Meat Supplies. Trade and Empire. Australia. New Zealand. Argentina. Brazil. Uruguay. Paraguay. Chile. Patagonia. United States. Canada. Mexico. Venezuela. British Guiana. Ecuador. Colombia. Honduras. Bolivia. Peru. Rhodesia. South Africa. Madagascar. Morocco. British East Africa. South-West Africa. Egypt and Soudan. Sheep in Egypt and Southern Palestine. Nigeria. Abyssinia, Somaliland and the Cameroons. China. Siberia. The Pacific Islands (Fiji, New Caledonia, Sandwich Group). Japan. Innovation in Mutton Stowage. Dipping Cattle. Cattle Tick Eradication. Dehorning Cattle. Live and Dressed Weights. Venison. Statistical Study of Body Weights, Gains and Measurements of Steers during the Fattening Period, with Tables. List of Meat Canning and Preserving Works in Australia, New Zealand, Argentina, Patagonia, Uruguay, Paraguay, Brazil, Venezuela.

International Language and Science. Considerations on the Introduction of an International Language into Science. Translated by **Prof. F. G. Donnan.** Demy 8vo. 2/- net.

Good Engineering Literature. **Harwood Frost, M.Am.Soc.M.E.** Crown 8vo. 4/6 net.

Life and Letters of Joseph Black, M.D. **Sir William Ramsay, K.C.B., F.R.S.** With an Introduction dealing with the Life and Work of Sir William Ramsay, by **F. G. Donnan, F.R.S.** Demy 8vo. 168 pages. 7 Illustrations. 6/6 net.

"A valuable contribution to scientific history."—*The Times.*

Britain's Heritage of Science. **Arthur Schuster, F.R.S., and Arthur E. Shipley, F.R.S.** Crown 8vo. 350 pages. 15 Portraits. 3rd Impression. 14/- net.

"This is the first attempt to give a comprehensive view of the history of British science in all its branches by men who occupy the position of authorities, and we congratulate them on the success of their efforts."—*British Medical Journal*.

From Newton to Einstein. Changing Conceptions of the Universe. **Benjamin Harrow, Ph.D.** Foolsap 8vo. Paper. 2/6 net.

Einstein's contributions to our ideas of time and space and to our knowledge of the universe in general must rank among the greatest achievements of the twentieth century. This little book gives a concise and lucid account of Einstein's discoveries, but since these are so largely dependent on the work of Newton and his successors, the first two chapters are devoted to the latter.

Panama. Its Creation, Destruction, and Resurrection. **Philippe Bunau-Varilla.** Royal 8vo. 12/6 net.

Messrs. Franklin and Charles' List of Publications

Practical Physics. Franklin, Crawford and Macnutt.

Vol. I. PRECISE MEASUREMENTS—MECHANICS AND HEAT. 9/- net.

Vol. II. ELEMENTARY AND ADVANCED MEASUREMENTS IN ELECTRICITY AND MAGNETISM. 9/- net.

Vol. III. PHOTOMETRY. EXPERIMENTS IN LIGHT & SOUND. 7/- net.

The Elements of Electrical Engineering. W. S. Franklin and William Esty.

Vol. I. DIRECT CURRENT MACHINES: ELECTRIC DISTRIBUTION AND LIGHTING. 28/- net.

Vol. II. ALTERNATING CURRENTS. 26/- net.

Advanced Theory of Electricity & Magnetism. W. S. Franklin and Barry MacNutt. 16/6 net.

Elements of Electricity and Magnetism. W. S. Franklin and Barry MacNutt. 12/- net.

The Elements of Electrical Engineering. W. S. Franklin. A Simplified Version of the "Elements" by Franklin and Esty (see above).

Vol. I. DIRECT AND ALTERNATING CURRENT MACHINES AND SYSTEMS. 28/- net.

Vol. II. ELECTRIC LIGHTING AND MISCELLANEOUS APPLICATIONS OF ELECTRICITY. 20/- net.

A Calendar of Leading Experiments. W. S. Franklin and Barry MacNutt. 16/6 net.

The Elements of Calculus. Franklin, MacNutt and Charles. 16/6 net.

Dynamo Laboratory Manual. W. S. Franklin & William Esty.

Vol. I. DIRECT-CURRENT STUDIES AND TESTS. 12/- net.

Elementary Electricity and Magnetism. **W. S. Franklin** and
Barry MacNutt. 9/- net.

Elementary Statics. **Franklin** and **MacNutt.** 4/- net.

Bill's School and Mine. A Collection of Essays by **W. S. Franklin.**
2nd Edn. 5/6 net.

Electric Waves. **W. S. Franklin.** 24/- net.

Lessons in Mechanics (1919). **W. S. Franklin** and **B. MacNutt.**
12/- net.

Lessons in Electricity and Magnetism (1919). **W. S.**
Franklin and **B. MacNutt.** 12/6 net.

INDEX

A

Adjustment of Observations by the Method of Least Squares, [29](#)

Advertising, A Study of a Modern Business Power, [55](#)

Aerial Flight, [20](#)

Agglutinants and Adhesives of all Kinds for all Purposes, [54](#)

Agriculture and Irrigation, [32](#), [58](#)

Air Navigation, A primer of, [19](#)

Air, Resistance of the, and Aviation, [19](#)

Aircraft in Warfare, [20](#)

Airscrews, [19](#)

Alternating Currents, [65](#)

Altitude and Health, [3](#)

Aluminium, The Metallurgy of, [8](#)

Arbitration Clause in Engineering and Building Contracts, [61](#)

Arc Lamps and Accessory Apparatus, [7](#)

Astronomy, [8](#)

Atom, The Universe and the, [48](#)

Atoms, [48](#)

B

Basic Open-Hearth Steel Process, The, [18](#)

Bells, Indicators, Telephones, Alarms, etc., [7](#)

Bill's School and Mine, [65](#)
Biology, Outlines of Evolutionary, [58](#)
Black, Joseph, M.D., Life and Letters of, [64](#)
Boiler Draught, [12](#)
Boiler Explosions, Collapses and Mishaps, [12](#)
Boilers, Marine Double-ended, [12](#), [21](#)
Book, The, [6](#)
Botany, [8](#)
Brass, The Metallurgy of, [9](#)
Britain's Heritage of Science, [64](#)
British in China, The, and Far Eastern Trade, [61](#)
British Railways, [63](#)
Building in London, [29](#), [61](#)

C

Calculus, An Elementary Treatise on the, [50](#)
Calculus, The, and its Applications, [50](#)
Calculus, The Elements of the, [65](#)
Calendar of Leading Experiments, A, [43](#), [65](#)
Cams, and the Principles of their Construction, [16](#)
Cement, Concrete, and Bricks, [8](#), [28](#), [43](#)
Cement Specifications, A Treatise on, [26](#), [27](#)
Cement Testing, Manual of, [26](#)
Centrifugal Pumps, [17](#)
Chadwick Library, The, [3](#)
Chemical Analysis, Public Health, [62](#)

Chemical Annual, Van Nostrand's, [49](#)
Chemical Calculations, [45](#)
Chemical Science, What Industry Owes to, [4](#), [48](#)
Chemical Theory and Calculations, [45](#)
Chemicals, The Manufacture of, by Electrolysis, [7](#), [39](#)
Chemistry, [8](#)
Chemistry, The, of the Coal Tar Dyes, [43](#)
Chemistry, The, of Dyeing and Bleaching of Vegetable Fibrous
Materials, [8](#), [44](#), [54](#)
Chemistry, A New Era in, [46](#)
Chemistry, The, of the Oil Industries, [8](#), [43](#), [51](#)
Chemistry, Old and New, Some Fundamental Principles of, [46](#)
Chemistry of Paints, The, and Paint Vehicles, [43](#)
Chemistry and Physics of Building Materials, Introduction to the, [6](#), [43](#)
Chemistry, The Profession of, [48](#)
Chemistry, The, of the Rubber Industry, [8](#), [43](#), [54](#)
Chemistry and Technology of Printing Inks, [42](#), [53](#)
Chemistry, Text-Book of Elementary, [46](#)
Circular-Arc Bow-Girder, A Study of the, [28](#)
Civil Engineers' Pocket-Book, [27](#)
Clouds, [61](#)
Coal, [5](#), [14](#)
Cold Storage, Heating and Ventilating on Board Ship, [21](#)
Colloidal and Crystalloidal State of Matter, [44](#)
Colloids, The, in Health and Disease, [3](#)
Colour and its Applications, [62](#)

Commerce and Industry, [56](#)
Commercial Paints and Painting, [6](#), [52](#)
Compressed Air. Its Production, Uses and Applications, [17](#)
Coniferous Trees, [60](#)
Contemporary Chemistry, [46](#)
Continuous Current Engineering, [37](#)
Conveyance and Distribution of Water for Water Supply, [22](#)
Cork, Its Origin and Industrial Uses, [54](#)
Corpuscular Theory of Matter, The, [47](#)
Cotton, [3](#), [54](#)
Cotton Seed Products, [52](#)
Curves of Life, The, [63](#)

D

Dairy Laboratory Guide, [59](#)
Designing, Theory and Practice of, [29](#)
Dictionary of British Scientific Instruments, [61](#)
Dictionary, Motor, [19](#), [61](#)
Dies, Their Construction and Use for the Modern Working of Sheet Metals, [16](#)
Diesel Engine Design, [10](#)
Diesel Engine, The, for Land and Marine Purposes, [10](#)
Direct and Alternating Current Machines and Systems, [65](#)
Direct and Alternating Current Testing, [38](#)
Direct Current Machines, [65](#)
Dock and Lock Machinery, [4](#), [16](#)

D.-S. Series of Technical Dictionaries, The, in Six Languages, [9](#)
Dust Destructors, Small, for Institutional and Trade Refuse, [31](#)
Dynamics of Mechanical Flight, [19](#)
Dynamo Laboratory Manual, [36](#), [65](#)
Dynamometers, [15](#)

E

Earth Slopes, Retaining Walls and Dams, Graphical Determination of
28
Earthwork in Railway Engineering, [4](#), [25](#)
Egg Farming, Commercial, [59](#)
Electric Arc Phenomena, [47](#)
Electric Cables and Networks, The Theory of, [37](#)
Electric Central Station Distribution Systems (American), [37](#)
Electric Cooking, Heating, Cleaning, etc., [34](#)
Electric Currents, The Propagation of, in Telephone and Telegraph
Conductors, [32](#), [36](#)
Electric Lamps, [5](#), [38](#), [39](#)
Electric Lighting and Miscellaneous Applications of Electricity, [65](#)
Electric Mechanism, [35](#)
Electric Mine Signalling Installations, [23](#), [34](#)
Electric Mining Installations, [7](#)
Electric Power, Application of, to Mines and Heavy Industries, [23](#), [33](#)
Electric Power and Traction, [5](#), [37](#)
Electric Railway Engineering, [37](#)
Electric Railways, Single-phase, [25](#), [33](#)
Electric Searchlight Projectors, The Range of, [38](#)

Electric Waves, [65](#)

Electric Welding and Welding Appliances, [5](#), [34](#)

Electric Wiremen's Manuals, [7](#), [38](#)

Electrical Conductivity, The, and Ionization Constants of Organic Compounds, [36](#)

Electrical Conductors, Calculation of, [35](#)

Electrical Engineering (D.-S. Dictionaries), [9](#)

Electrical Engineering, The Elements of, [36](#), [65](#)

Electrical Engineering, Heavy, [37](#)

Electrical Engineering, Essentials of, [38](#)

Electrical Engineer's Pocket Book (Foster's), [36](#)

Electrical Machinery, Testing, [37](#)

Electrical Measuring Instruments, Industrial, [35](#)

Electrical Nature of Matter and Radio-Activity, [47](#)

Electricity and Magnetism, Elementary, [36](#), [65](#)

Electricity and Magnetism, Elementary and Advanced Measurements in, [48](#), [65](#)

Electricity and Magnetism, Advanced Theory of, [36](#), [65](#)

Electricity and Magnetism, Elements of, [36](#), [65](#)

Electricity and Magnetism, Lessons in, [65](#)

Electricity and Matter, [48](#)

Electricity, A Text-Book of Applied, [36](#)

Electro-Chemistry, A Treatise of, [7](#)

Electro-Metallurgy, [6](#), [39](#)

Electrolysis, The Manufacture of Chemicals by, [7](#), [39](#), [42](#)

Elements of Machinery, The (D.-S. Dictionaries), [9](#)

Energy Diagram, The, for Gas, [11](#)

Engine Tests and Boiler Efficiencies, [12](#)
“Engineer Library,” The, [4](#)
Engineering Literature, Good, [64](#)
Engines and Auxiliaries, The Design of Marine, [20](#)
Entropy, [17](#), [39](#)
Estate Economics, [60](#)
European Animals, [57](#)
Extinct Animals, [57](#)

F

Farm Animals, [59](#)
Farming, The First Book of, [59](#)
Field Engineer, The, [25](#)
Floating Bodies, Stability and Equilibrium of, [20](#), [22](#)
Flower and the Bee, The, [58](#)
Flying Machine, The, from an Engineering Point of View, [20](#)
Food Adulterants, Detection of the Common, [43](#)
Forecasting Weather, [61](#)
Forests, Woods and Trees in Relation to Hygiene, [3](#), [60](#)
Foundations and Machinery Fixing, [7](#), [28](#)
Foundry Practice, General, [6](#), [18](#)
Frozen Meat Trade, History of the, [54](#)
Fuel, An Introduction to the Study of, [8](#), [14](#)
Fuel Consumption, Domestic, [3](#), [14](#)
Fuel, Water and Gas Analysis for Steam Users, [13](#)
Fuels, Low Grade and Waste, for Power Generation, [14](#)

Future Citizen, The, and His Mother, [3](#)

G

Gas Engine, The, [6](#), [11](#)

Gas Engine Construction, [11](#)

Gas Engine Design, [11](#)

Gas-Engineering Practice (American), [11](#)

Gas Engines, High Power, [11](#)

Gas, Gasoline, and Oil Engines, [11](#)

Gas, Town, for Lighting and Heating, [5](#), [39](#)

Gas Turbine, The, [11](#)

Gas Works Plant and Machinery, British Progress in, [11](#)

Geographic Environment, Influences of, [58](#)

Glasgow Text-Books of Civil Engineering, The, [4](#)

Glass Manufacture, [6](#), [53](#)

Glass Processes, Decorative, [5](#), [53](#)

Glues and Gelatine, [54](#)

Governors and the Governing of Prime Movers, [10](#), [16](#)

Graphic Statics, The Elements of, and of General Graphic Methods, [21](#)

Gyroscopic Compass, The, [5](#), [20](#)

H

Health in Camp, [3](#)

Heat Power-Plants, The Economic and Commercial Theory of, [17](#)

Heat Transmission, [40](#)

Hoisting and Conveying Machinery (D.-S. Dictionaries), [9](#)

Hydraulic Tables for the Flow of Water in Circular Pipes, [23](#)
Hydraulics and its Applications, [22](#)
Hydraulics, Text-Book of, [22](#)
Hydraulics of Pipe Lines, [4](#), [22](#)
Hydroelectric Developments and Engineering, [23](#), [37](#)
Hydrogenation of Oils, The, [43](#)

I

Identification of Organic Compounds, The, [46](#)
India-Rubber and its Manufacture, [6](#), [54](#)
Indian Trees, [60](#)
Indicators, The Theory and Use of, [45](#)
Industrial Accidents and their Compensation, [61](#)
Industrial Chemistry, [41](#)
Industrial Chemistry, Elements of, [42](#)
Industrial Chemistry, Outlines of, [8](#)
Industrial Chemistry, Laboratory Guide of, [42](#)
Industrial Engineering: Its Present Position and Post-War Outlook, [63](#)
Industrial Peace, Terms of, [63](#)
Industry, What it owes to Chemical Science, [4](#), [48](#)
Inorganic Chemistry, Practical Methods of, [46](#)
Insect Pests and Plant Diseases, [59](#)
Integration by Trigonometric and Imaginary Substitution, [50](#)
Internal Combustion Engine, The, [10](#)
Internal Combustion Engine, A Primer of the, [10](#)
Internal Combustion Engines (D.-S. Dictionaries), [9](#)

International Language and Science, [64](#)
Ionization of Gases by Collision, The Theory of, [47](#)
Iron and Steel, [8](#), [18](#)
Iron and Steel Construction, Electro-Thermal Methods of, [18](#)
Iron and Steel, Forging of, [16](#), [18](#)
Irrigation: Its Principles and Practice as a Branch of Engineering, [32](#)
Irrigation Works, [32](#)
Irrigation Works, Notes on, [32](#)
Irrigation Works, The Practical Design of, [32](#)

L

Laboratory and Factory Tests in Electrical Engineering, [37](#)
Labour, The Claims of, and of Capital, [63](#)
Law Affecting Engineers, The, [57](#), [61](#)
Leather, The Manufacture of, [53](#)
Lettering, [57](#)
Life in America, Distribution and Origin of, [57](#)
Life Histories of Northern Animals, [58](#)
Light and Shade and their Applications, [62](#)
Limitations of Science, The, [63](#)
Linen, [3](#)
Linseed Oil and other Seed Oils, [51](#)
Liquid Air and the Liquefaction of Gases, [43](#)
Liquid Fuel and its Apparatus, [14](#)
Liquid Fuel and its Combustion, [14](#)
Liquid and Gaseous Fuels, [5](#), [14](#)

Living Plant, The, [58](#)

Locomotive Valves and Valve Gears, [24](#)

Locomotives, Oil Fuel Equipment for, [13](#), [24](#)

M

Machine Tools (D.-S. Dictionaries), [9](#)

Malleable Cast Iron, [18](#)

Marine Boilers, Design of, [12](#), [21](#)

Masonry Applied to Civil Engineering, [28](#)

Mathematics, [8](#)

Mathematics for the Practical Man, [51](#)

Measurements, Theory of, [46](#)

Mechanical Appliances, [15](#)

Mechanical Movements, Powers, Devices and Appliances, [15](#)

Mechanical Technology, [15](#)

Mechanics, [8](#)

Mechanics and Heat, Precise Measurements, [48](#), [65](#)

Mechanics, Lessons in, [65](#)

Mechanic's Handbook, The Practical, [51](#), [56](#)

Mechanics of Materials, The Elements of, [15](#)

Mechanics, The Theory and Practice of, [15](#)

Metallurgical Analysis, Methods in, [49](#)

Metallurgy: A Series, [9](#)

Metallurgy (D.-S. Dictionaries), [9](#)

Microscopy, Principles of, [49](#)

Mineral and Aerated Waters, and the Machinery for their Manufacture, [53](#)

Mining and Mine Ventilation, Physics and Chemistry of, [23](#)

Motion Study, [56](#)

Motor-Car Engineering, Text-Book of, [19](#)

Motor Dictionary, Henslowe's, [19](#), [61](#)

Motor Vehicles (D.-S. Dictionaries), [9](#)

Motors, Secondary Batteries, Measuring Instruments and Switchgear, [7](#)

Municipal Engineering, British Progress in, [31](#)

Municipal and Sanitary Engineering, The Encyclopædia of, [30](#)

N

Natural Rock Asphalts and Bitumens, [62](#)

Naval Machinery, Handbook for the Care and Operation of, [21](#), [57](#)

Newton to Einstein, From, [64](#)

Nitrocellulose Industry, The, [44](#), [54](#)

Nitrogen Industry, The, [3](#)

O

Oceanography, A Text-Book of, [63](#)

Oil Engines, Hot Bulb, and Suitable Vessels, [10](#), [20](#)

Oil Firing for Kitchen Ranges, [14](#)

Oils, Gums, Waxes and Allied Substances, [54](#)

Overhead Transmission Lines and Distributing Circuits, [34](#)

Ozone, [7](#), [39](#), [49](#)

P

Painting, Materials for Permanent, [44](#), [53](#)

Panama: Its Creation, Destruction, and Resurrection, [64](#)

Paper Making and its Machinery, [5](#), [52](#)

Paper, The Manufacture of, [5](#), [52](#)

Patent Laws, Foreign and Colonial, [61](#)

Patents, Designs and Trade Marks, [61](#)

Patents, The Law and Commercial Uses of, [6](#)

Perspective, [62](#)

Photography, [6](#)

Photography: Its Principles and Applications, [55](#)

Photometry, Experiments in Light and Sound, [48](#), [65](#)

Physical Chemistry, Exercises in, [48](#)

Physical Chemistry, Problems in, [45](#)

Physical Laboratory Experiments for Engineering Students, [47](#)

Physical Metallurgy, An Introduction to the Study of, [9](#), [17](#), [46](#), [49](#)

Physico-Chemical Theory, The Application of, to Technical Processes and Manufacturing Methods, [42](#), [51](#)

Physics, Practical, [48](#), [65](#)

Physics, A Text-Book of, [47](#)

Plane Trigonometry, Elementary, [50](#)

Plants and Their Uses, [58](#)

Plant Physiology and Ecology, [58](#)

Potter's Craft, The, [54](#)

Powdered Coal as a Fuel, [13](#)

Power, Application of the Principles of, to Road Transport, [27](#)

Power, Natural Sources of, [6](#), [17](#)
Power Plants, Steam-Electric, [37](#)
Practical Manuals for Practical Men, [7](#)
Precious Metals, The, [5](#)
Precious Stones, [6](#), [18](#), [55](#)
Precision Grinding, [16](#)
Pricing of Quantities, [29](#)
Proportion, The Theory of, [50](#)
Public Health Chemical Analysis, [62](#)
Pumps and Ejectors, The Efficiency of, [17](#), [30](#)
Purchasing: Its Economic Aspects and Proper Methods, [55](#)

Q

Quantitative Analysis, Principles of, [44](#)
Quaternions, as the Result of Algebraic Operations, [50](#)

R

Radio-Active Transformations, [47](#)
Radio-Activity and Geology, [47](#)
Radiographic Technique, [33](#)
Radio-Telegraphy, [6](#), [33](#)
Railway Construction and Operation (D.-S. Dictionaries), [9](#)
Railway Locomotive, The, [5](#), [25](#)
Railway Maintenance Engineering, [24](#)
Railway Rolling Stock (D.-S. Dictionaries), [9](#)
Railway Signal Engineering (Mechanical), [4](#), [25](#)

Railway Spiral, The Practical, [24](#)
Railway Terms, Spanish-English, English-Spanish, Dictionary of, [25](#)
Rainfall, Reservoirs and Water Supply, [3](#), [29](#), [30](#), [31](#)
Reagent Chemicals, Standards and Tests For, [40](#)
Refuse Disposal and Power Production, [30](#)
Reinforced Concrete Bridges, [25](#), [26](#)
Reinforced Concrete Compression Member Diagram, [27](#)
Reinforced Concrete, A Concise Treatise on, [26](#)
Reinforced Concrete Diagrams, [26](#)
Reinforced Concrete, A Manual of, [26](#)
Reinforced Concrete, Properties and Design of, [26](#)
Reinforced Concrete Railway Structures, [4](#), [25](#), [26](#), [28](#)
Reinforced Concrete in Sub- and Superstructure (D.-S. Dictionaries), [9](#)

S

Sanitary Engineering, Modern, [4](#), [31](#)
Sea Fisheries, The, [63](#)
Searchlights: Their Theory, Construction and Application, [38](#)
Seasonal Trades, [55](#), [63](#)
Sewage Disposal Work: Design, Construction and Maintenance, [30](#)
Science, The Seven Follies of, [63](#)
Scientific Management, Primer of, [56](#)
Shipbuilding Industry, [21](#)
Ship Form, Resistance, and Screw Propulsion, [21](#)
Ship Wiring and Fitting, [7](#)
Shop Kinks, [16](#)

Silicates, The, in Chemistry and Commerce, [41](#)

Simple Jewellery, [55](#)

Single-Phase Electric Railways, [33](#)

Single-Phase Commutator Motors, [35](#)

Sizing, Materials Used in, [54](#), [56](#)

Slide Rule, Solution of Railroad Problems by the, [24](#)

Soft Soldering, Hard Soldering, and Brazing, [62](#)

Soils and Manures, [5](#), [59](#)

Solenoids, Electromagnets and Electromagnetic Windings, [35](#)

Solution, The Nature of, [45](#)

Solvents, Oils, Gums, Waxes and Allied Substances, [54](#)

Staple Trades and Industries Series, [3](#)

Statical Calculations, Reference Book for, [56](#)

Statics, Elementary, [50](#), [65](#)

Steam Boilers, [13](#)

Steam Boilers and Steam Engines (D.-S. Dictionaries), [9](#)

Steam Engine, A Manual of the High-Speed, [12](#)

Steam Engine, The Modern, [12](#)

Steam Engine, The, and Turbine, [12](#)

Steam, Experimental Researches on the Flow of, through Nozzles, [13](#)

Steam Pipes, [13](#)

Steam Power Plants, Test Methods for, [12](#)

Steam Tables, The New, [13](#), [56](#)

Steel, Electro-Metallurgy of, [49](#)

Steel, Hardening, Tempering, Annealing and Forging of, [18](#)

Steel Bridges, The Design of Simple, [25](#), [27](#)

Steel-Framed Sheds, Practical Design of, [28](#)
Steel Rails: Their History, Properties, Strength and Manufacture, [24](#)
Storing: Its Economic Aspects and Proper Methods, [55](#)
Strain and the Refractory Metals, Metallurgy of, [9](#)
Structural Design, The Elements of, [28](#)
Structural Geology, [62](#)
Surveying and Field Work, [29](#)
Surveying, Practical, [30](#)
Switchgear and the Control of the Electric Light Power Circuits, [7](#)
Switchgear Design, High and Low Tension, [35](#)
Switches and Switchgear, [35](#)

T

Telegraph Engineering, [32](#)
Telegraphic Transmission of Photographs, The, [33](#)
Telegraphy, Aeronautics and War, [33](#)
Telephone and Telegraph Wires, The Propagation of Electric Currents in, [32](#)
Temperatures, The Measurement of Steady and Fluctuating, [40](#)
Testing, A Handbook of, [15](#)
Testing and Localizing Faults, [7](#)
Thermodynamics, Applied, for Engineers, [40](#)
Thermodynamics, Experimental and Theoretical Applications of, to Chemistry, [40](#), [42](#)
Thermodynamics, Principles of, [40](#)
Thermodynamics, Technical, [40](#)
Thermodynamics, A Text-Book of, [40](#), [46](#)

Thresholds of Science, [8](#)

Timber, [5](#), [29](#)

Time and Clocks, [62](#)

Toll Telephone Practice, [33](#)

Town Planning in Madras, [31](#), [62](#)

The Transition Spiral, [24](#), [30](#)

Transmission Line Formulas, [34](#)

Trinitrotoluenes and Mono- and Dinitrotoluenes: Their Manufacture and Properties, [44](#)

Turbine Practice, Modern, and Water Power Plants, [23](#)

Turbines Applied to Marine Propulsion, [21](#)

U

Universe, The, and the Atom, [48](#)

Utility Ducks and Geese, [59](#)

V

Vapours for Heat Engines, [42](#)

Vegetable Oils, The Production and Treatment of, [4](#), [52](#)

Ventilation, Some Modern Methods of, [27](#)

W

Water, Conveyance and Distribution of, for Water Supply, [30](#)

Water Hammer in Hydraulic Pipe Lines, [22](#)

Water Pipe and Sewage Discharge Diagrams, [23](#), [31](#)

Water Purification and Sewage Disposal, [30](#), [31](#)

Water: Its Purification and Use in the Industries, [31](#), [62](#)

Water Softening and Treatment, [16](#)

Welding and Cutting of Metals by the Aid of Gases or Electricity, [6](#), [18](#)

“Westminster” Series, The, [5](#), [6](#)

Wiring of Buildings, The Internal, [38](#)

Wiring, Mill and Factory, [7](#)

Wood Pulp, [6](#), [52](#)

Wool, [3](#), [54](#)

World’s Meat Future, The, [64](#)

Z

Zinc, The Metallurgy of, [9](#)

Zoology, [8](#)

Zoology and Entomology, Economic, [57](#)

The Mayflower Press, Plymouth, England. William Brendon & Son, Ltd.

*** END OF THE PROJECT GUTENBERG EBOOK A CATALOGUE OF
TECHNICAL AND SCIENTIFIC BOOKS PUBLISHED BY
CONSTABLE & COMPANY LTD ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.